



BASE24-atm[®]

Diebold 910/911/912 Device Support Manual

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Print Record

08/96, Version 2 (Reissue)

12/93, Version 1 (New)

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Preface

The ***BASE24-atm Diebold 910/911/912 Device Support Manual*** provides an overview of the Diebold 910/911 and 912 Device Handler processes and Diebold-related software options required by BASE24-atm. This manual discusses the interaction between BASE24 and the Diebold 910, 911, and 912 ATMs. It also examines the role of BASE24 in the features, specifications, and functions of the Diebold 910, 911, and 912 ATMs.

Audience

ACI prepared this publication for distribution to institution personnel responsible for updating and maintaining Diebold 910, 911, and 912 ATMs for use with BASE24-atm.

Prerequisites

This manual discusses a number of general topics relating to Diebold 910, 911, and 912 ATMs. These discussions have been included primarily to provide a stronger insight into the way BASE24-atm handles Diebold 910, 911, and 912 ATMs. This manual is not intended as a substitute for the vendor literature relating to Diebold 910, 911, and 912 ATMs. It is recommended that individuals making decisions on BASE24-atm Diebold 910, 911, and 912 ATM processing be familiar with the vendor literature available for the type of Diebold 910, 911, and 912 ATMs they are using.

Additional Documentation

The BASE24 documentation set is arranged so that each BASE24 manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic is summarized and the reader is directed to the other manual for additional information. Information has been arranged in this manner to be more efficient for readers who do not need the additional detail and at the same time provide the source for readers who require the additional information.

This manual contains references to the following BASE24 publications:

- The **BASE24 Device Control Manual** includes instructions for such actions as starting and stopping the ATM, downloading information to the ATM, and performing status inquiries.
- The **BASE24 Text Command Reference Manual** describes the network control facilities.
- The **BASE24 CRT Access Manual** describes how to access and use the BASE24 files maintenance menus and screens.
- The **BASE24-atm Diebold PAMS Support Manual** contains information on BASE24 support of Performance Analysis Management System (PAMS).
- The **BASE24 Transaction Security Manual** contains complete information on BASE24 transaction security options and message authentication codes (MACs).
- The **BASE24-atm Files Maintenance Manual** describes the Terminal Data File (TDF) and the Receipt File (RCPT).
- The **BASE24 Base Files Maintenance Manual** describes the Institution Definition File (IDF) and the Surcharge File (SURF).
- The **BASE24-atm Settlement and Reporting Manual** contains a complete description of BASE24-atm settlement and cutover process and reports generated by BASE24.
- The **BASE24-atm Transaction Processing Manual** describes the BASE24-atm transactions.
- The **BASE24 Logical Network Configuration File Manual** contains detailed information on the assigns and params in the Logical Network Configuration File (LCONF).
- The **BASE24 Log Message Reference Manual** contains a complete explanation of log messages and information on how to access the documentation for log messages generated by the Diebold 910/911 and 912 Device Handler processes.
- The **BASE24-atm Tokens Manual** describes the tokens which are appended to the Standard Transaction Message (STM).
- The **XPNET BASE24 Configuration Guidelines** describes the Log Message Configuration File (LMCF) and setting up the source declarations for each terminal in the XPNET node.

In addition, this manual contains references to the following vendor publications:

- The *Guardian User's Guide* contains information on the RUN command.
- The *Diebold TABS Manual, #386* and the *MDS Enhanced Character Supplement to the TABS 912/922 Programming Manual* contain further information on the enhanced display feature.
- The *Diebold Data Security Procedures and Reference Manual, October 1983* contains information on setting up the Configuration File to support local PIN verification.

Software

This manual documents standard processing for BASE24 Release 5.1.

Customer-specific code modifications (that is, RPQs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Manual Summary

The following is a summary of the contents of this manual.

“Conventions Used in this Manual” follows this preface and describes notation and documentation conventions necessary to understand the information in the manual.

Section 1, “Introduction,” introduces the reader to the Diebold 910, 911, and 912 ATM models that are available and describes how BASE24-atm provides support for Diebold 910, 911, and 912 ATMs in the following areas: ATM features, protocols used, Device Handler process issues, and financial and administrative transactions.

Section 2, “ATM Configuration,” contains information about Diebold 910, 911, and 912 ATM configuration requirements, Diebold 910/911 and 912 Configuration Files, and the BASE24 files used in configuring Diebold 910, 911, and 912 ATMs.

Section 3, “BASE24-atm Device Handler Process,” contains an overview of the BASE24-atm Diebold 910/911 and 912 Device Handler processes, descriptions of the processing steps the Device Handler process takes when it is first started, an

overview of the network commands that can be sent to the Device Handler process, and information on how to access the log messages generated by the Device Handler process.

Section 4, “Configuration File Maintenance,” describes the Diebold 910/911 and 912 Configuration Files (CNFG910 and CNFG912) maintenance that can be performed using BASE24-atm. Also, it includes the various screens used to change information in the Configuration File.

Section 5, “ATM Printer Support,” describes printer support for Diebold 910, 911, and 912 ATMs. Examples of receipts generated at a Diebold 910, 911, and 912 ATM are shown as they would appear to the cardholder. These receipts are generated when a financial transaction is performed, and when an administrative transaction is performed.

This section also describes the configurable receipt option. This option enables an institution to establish ATM receipt formats that are unique to the institution.

Appendix A, “Philips Enhanced Display Screens,” contains documentation for the enhanced display configuration screens for Diebold 912-series ATMs equipped with the Philips enhanced character generator programmable read only memory (PROM).

Publication Identification

Two entries appearing at the bottom of each page uniquely identify this BASE24 publication. The publication number (for example, AT-DA020-01 for the ***BASE24-atm Diebold 910/911/912 Device Support Manual***) appears on every page to assist readers in applying updates to the appropriate manual. The publication date (for example, 08/96 for August, 1996) identifies the last time material on a particular page has been revised. This date can be used to verify that the reader has the most current information and is particularly helpful when dealing with HELP24 to answer operating questions.

The print record, located on the page immediately following the title page, is also helpful when verifying that a manual contains the most current information. An entry is added to the print record each time a BASE24 publication is modified in any way. Each print record entry includes the date of the change, a unique version number, and the type of change (new, update, or reissue). Version 1 is assigned to a *new* manual. Later versions can be updates or reissues, depending on the changes being made. An *update* typically involves changes on a small number of pages throughout the manual. A *reissue* occurs when all pages of the manual are replaced because changes affect a large number of pages.

Conventions Used in this Manual

This section explains syntax notation for entering commands and how fields are documented in this manual. It also explains a change in Nucleus process terminology.

Syntax Notation

Syntax descriptions in this manual use several symbols and conventions to denote how commands must be entered. These conventions are described in the table below.

Notation	Meaning
UPPERCASE LETTERS	Indicate key words and literals that must be entered exactly as shown.
<i>lowercase italics</i>	Identify variables that you must provide.
Brackets []	Enclose optional syntax items that you can enter in the command.
Braces { }	Enclose a group of items from which you are required to choose one item. The items are separated within the braces by a vertical bar ().
Vertical bar	Separates alternatives in a list of items.
Ellipsis ...	Indicates the immediately preceding syntactical item can occur one or more times.
Punctuation	Must be entered as shown. This includes parentheses, commas, semicolons, and other symbols not previously described.

Notation	Meaning
Spaces	Are required as delimiters wherever it would be otherwise impossible to discriminate between adjacent words or symbols. You can insert additional spaces between any adjacent words or symbols, but not within a word or multiple character symbol.
Indented lines	Indicates a continuation of the preceding line of syntax. If the syntax of a command is too long to fit on a single line, each continuation line is indented.

Field Descriptions

Each field appearing on a file maintenance screen is listed by name and then described. Field descriptions briefly summarize the contents, purpose, and permissible values, as shown in the following example taken from the Diebold Configuration File.

Example

Screen Number — A number between 000 and 511 that is associated with the desired screen.

Field Length: 3 numeric characters
Required Field: Yes
Default Value: 000

Explanation

Each field description is completed by one or more of the following items of information:

Item	Description
Example	Demonstrates a possible entry for the field to further clarify the value or values that can be entered.

Item	Description
Field Length	Specifies the size of the field and the type of characters that you can enter. This length refers to the field and valid values for the file maintenance screen, not the field in the DDLs. Possible values are alphabetic (A–Z), alphanumeric (all characters), hexadecimal (0–9, A–F), and numeric (0–9). When you cannot modify a field value, the field length is System-protected.
Occurs	Indicates the number of times the field can be displayed on the screen. This information is given only if the field can be displayed multiple times.
Required Field	Specifies whether you have to enter a value in the field. Possible values are Yes and No. Some fields are required only under certain conditions. In this case, the entry is Yes, followed by the conditions that determine whether the field is required.
Default Value	Specifies the value that is automatically placed in the field when the screen is first displayed or when function key F8 is pressed to clear the screen.

Unlabeled Fields on Screens

Angle brackets (< >) indicate an unlabeled field on a screen. An unlabeled field is a field that is present on a screen but is not preceded by an identifying literal label. For the purposes of documenting the field, a label has been assigned and appears inside the angle brackets. A multiple line unlabeled field is displayed as a shaded area and also has a label in angle brackets.

In the field descriptions for the screen, the unlabeled field appears according to its place on the screen and is identified by the same label.

Nucleus Process Terminology Change

There is one process that is the hub of an XPNET node. It manages message queues; controls all lines, stations, and processes in the system; and serves as an interprocess interface. Throughout this manual, this process is referred to as the XPNET process.

In other manuals, you may see references to the Nucleus process as performing these same functions. These two terms (Nucleus process and XPNET process) should be viewed as synonymous. The Nucleus process and the XPNET process perform the same functions and processing. As manuals are reissued, references to the Nucleus process will be replaced with references to the XPNET process.

Section 1

Introduction

This section provides an introduction to BASE24-atm support of Diebold 910, 911, and 912 ATMs. It contains information about the following items:

- Diebold 910, 911, and 912 models supported by BASE24-atm
- Diebold 910, 911, and 912 features supported by BASE24-atm
- Protocols supported by BASE24-atm for Diebold 910, 911, and 912 ATMs
- Transaction security
- Cutover and settlement
- Financial and administrative transactions supported by BASE24-atm for the Diebold 910, 911, and 912 ATMs

Diebold TABS 9000 Models

BASE24-atm supports three models of Diebold TABS (Total Automatic Banking System) 9000 terminals: the 910, the 911, and the 912.

The Diebold TABS 910 is a full-service terminal that allows cardholders to perform routine banking transactions without direct assistance from bank personnel. The 910 terminals are available in two models: the through-the-wall drive-up model and the through-the-wall walk-up model.

The Diebold TABS 911 terminal is also available in both of the through-the-wall models plus four additional models: the free-standing lobby model, the in-counter lobby model, and two walk-up, stand-alone models designed for interior use. The model 911 performs the same transactions as the 910, but the 911 includes the Total Cash Control System (TCCS). The TCCS uses fully secured, locked cassettes for currency, captured cards, diverted currency, and the depository. The TCCS makes third-party servicing feasible.

The Diebold TABS model 912 features the following as standard equipment: the ability to display 64 ASCII characters plus 34 additional special characters; a level 2 card reader that can pass track 2 information to BASE24; card capturing capability and two-denomination bill dispensing from cassettes capability. Optionally, some of the features include enhanced pixel control display; journal print capability; tracks 1 and 3 reading with track 3 write capability; four-denomination bill dispensing and four document cassettes for contract servicing of the ATMs. Diebold 912/922 series are ATMs or Cash Dispensing, respectively, with walk-up and lobby configurations available in either series.

Diebold TABS 9000 terminal models are described below.

- The through-the-wall drive-up model meets the applicable requirements of Underwriters Laboratories (UL) 291 for a drive-up installation.
- The through-the-wall walk-up model meets the applicable requirements of Underwriters Laboratories (UL) specification 291 and the Bank Protection Act for exterior wall installation.
- The free-standing lobby model has side panels. With the optional inner door, this model meets UL specification 291 requirements for a lobby unit.
- The in-counter lobby model, without sidepanels, is suited for in-the-counter use.
- The stand-alone model I is a walk-up model that meets the applicable requirements of UL specification 291 and the Bank Protection Act for indoor installation in retail stores.

- The stand-alone model II is a model that is identical to the stand-alone model I, without the delivery and depository doors. This model meets the applicable requirements of UL specification 291 and the Bank Protection Act for indoor installation in retail stores.

BASE24-atm Support

BASE24-atm provides the following support specifically tailored to the Diebold 910, 911, and 912 terminals:

- Supported features
- Supported protocols
- Transaction security

Supported Features

Diebold offers a variety of options for the 910, 911, and 912 terminals, including alarm options, PIN-handling techniques, and hardware features, such as a camera, additional fascia, and an auxiliary printer. No matter which options are attached to the Diebold 910, 911, or 912 terminal, BASE24 is used as a processor for transactions originating at the ATM.

Several of the options available through Diebold are required in order for BASE24 to process transactions. These include the Dual Denomination option and the Track II Read option.

The following Diebold 910, 911, and 912 features are supported by BASE24-atm:

- Card reader
- Envelope depository
- DES PIN encryption
- ANSI X9.8 standard PIN block format
- Downloading Configuration Files
- Journal printer
- Receipt printer
- Delivery area sensor check (Diebold 912 only)
- Local PIN verification

Note: BASE24 supports the use of the motorized card reader. The motorized card reader receives cards in a right-handed, face up presentation and can be programmed to capture cards for security purposes.

Supported Protocols

BASE24-atm can support communications with Diebold terminals using the following protocols:

- Burroughs Poll/Select Asynchronous protocol
- IBM 3275 Bisynchronous ASCII protocol
- IBM 3275 Bisynchronous EBCDIC protocol
- IBM SDLC/SNA protocol
- SNAX/HLS protocol

BASE24-atm Diebold PAMS Support

The Diebold Performance Analysis Management System (PAMS) is supported by BASE24 as an add-on module for Diebold 910/911 and 912 Device Handler processes. PAMS allows institutions to collect statistical information for their Diebold devices. The information is stored in the Diebold Hardware Status File (DHSF) which can be extracted and written to a tape in a format specified by Diebold. The Diebold Corporation analyzes and reports the statistical performance information for institutions to help them achieve maximum performance from their Diebold terminals. For complete information on BASE24 support of PAMS, refer to the ***BASE24-atm Diebold PAMS Support Manual***.

Transaction Security

BASE24-atm offers transaction security options to protect an institution and its cardholders from fraud and includes such functions as PIN encryption, PIN verification, and support for message authentication.

BASE24-atm also supports the PIN Change transaction for the Diebold 910, 911, and 912 ATMs which enables a cardholder to perform a PIN change at an ATM. However, if local PIN verification is used, the PIN Change transaction is not supported.

BASE24-atm supports transaction security via both hardware and software. Hardware security options include interfaces to various models of Atalla and Racal-Guardata security hardware. Software security options include several methods of PIN encryption and PIN verification.

The BASE24-atm Diebold 910/911 and 912 Device Handler processes are responsible for the interface between BASE24 and the Diebold 910, 911, and 912 ATMs attached to them. In this capacity, the Diebold 910/911 and 912 Device Handler processes are responsible for PIN encryption between BASE24 and the Diebold 910, 911, and 912 terminals. Based on a combination of device capabilities and institution preferences, the Device Handler process passes or decrypts PINs upon entry into the system.

For complete information on BASE24 transaction security options, refer to the ***BASE24 Transaction Security Manual***.

Local PIN Verification

BASE24-atm supports local PIN verification in which Diebold 910, 911, and 912 devices, which have PIN verification capability, perform PIN verification at the device for financial and administrative transactions without the involvement of the BASE24 network.

To perform local PIN verification, the Diebold 910, 911, and 912 terminal must be equipped with the proper hardware. Only those PINs whose offset is encoded on the card can be verified at the ATM.

You must consider which cards can be verified at the ATM because some interchanges do not support a PIN preverified at the ATM prior to transmission to the host.

BASE24-atm does not keep track of the number of bad PIN attempts whenever the ATM performs PIN verification.

Local PIN verification requires a setting in the BASE24-atm Terminal Data File (TDF) as well as updates to the financial institution table (FIT) and various state tables in the Diebold 910/911 and 912 Configuration Files (CNFG910 or CNFG912). The Configuration File must be downloaded after changes are made to implement local PIN verification. Refer to section 4 for complete information on how to access the CNFG910 or CNFG912 using the facilities of BASE24. Refer to the ***Diebold Data Security Procedures and Reference Manual, October 1983*** for complete information on setting up the Configuration File to support local PIN verification.

Message Authentication Code

Message authentication ensures that transaction messages are received exactly as created by the legitimate originator. Message authentication protects financial transaction messages against accidental or deliberate alteration.

The message authentication code (MAC) is generated by the originator of the message, based upon message elements identified in advance by the originator and recipient, and included with the message.

BASE24-atm supports the generation and verification of MAC values in hardware. Full message authentication is supported for the following message types:

- Consumer Request (transaction request)
- Function Command (transaction response)
- Write Command I (download of state data)
- Write Command VI (download of FIT data)

Message authentication is supported by field settings in the device-specific portion of the BASE24-atm Terminal Data File (TDF). For complete descriptions of these fields, refer to the ***BASE24-atm Files Maintenance Manual***. For complete information on message authentication for Diebold 910, 911, and 912 terminals, refer to the appropriate Diebold vendor documentation. BASE24-atm supports transaction security via both hardware and software. Hardware security options include interfaces to various models of Atalla and Racal-Guardata security hardware. Software security options include several methods of PIN encryption and PIN verification.

For complete information on using message authentication codes (MACs), refer to the ***BASE24 Transaction Security Manual***.

Cutover and Settlement

The role of BASE24-atm in settlement procedures consists of providing you with detailed transaction reports that reflect cardholder and terminal activity on both a daily and monthly basis. BASE24-atm defines the business day, captures and keeps track of transaction totals for each day, provides a daily file of transactions for processing by a host, and reports the funds due from and owed to the various participants in the logical network for balancing purposes.

A term used in connection with the BASE24-atm settlement process is cutover. Cutover refers to closing the current posting day and opening the next. It is the point in time when transactions stop being logged to the business for the current day and start being logged to the business for the next day by each terminal in the institution network.

For a complete description of BASE24-atm settlement and cutover, refer to the ***BASE24-atm Settlement and Reporting Manual***.

Transaction Support

BASE24-atm supports a number of financial and administrative transactions for Diebold 910, 911, and 912 ATMs. They are listed below.

Financial Transactions

The following financial transactions are supported by BASE24-atm for the Diebold 910, 911, and 912 ATMs:

- Withdrawal from checking account
- Withdrawal from savings account
- Cash advance from credit account
- Fast cash
- Deposit to checking account
- Deposit to savings account
- Transfer from checking to savings
- Transfer from savings to checking
- Transfer from credit account to checking
- Transfer from checking to checking
- Transfer from savings to savings
- PIN change at the host
- Payment enclosed
- Payment from checking to credit account
- Payment from savings to credit account
- Balance inquiry from checking
- Balance inquiry from savings
- Inquiry into credit account availability
- Message from customer to financial institution
- PIN change at the ATM
- Log only #1
- Log only #2

- Log only #3
- Log only #4

For more information about BASE24-atm transactions, refer to the ***BASE24-atm Transaction Processing Manual***.

Transaction Receipt Printing

All of the transactions listed above can have receipts printed for the cardholder's records. There are several ways that you can determine whether the receipt will print, and if so, how the receipt will look. Refer to section 5 for a further discussion of these options.

Optional Receipts

You can determine whether a receipt will be printed by transaction type. You also can allow the cardholder to determine if a receipt will be printed on all transactions except for the administrative transactions. This second option is available only for Diebold 910 and 911 terminals.

Configurable Receipts

The BASE24-atm product comes with standard formats, or templates, for how financial and administrative receipts will look. You can use these templates, or modify them to print the information that you want the cardholder to receive. Header and trailer templates allow you to customize the receipt to print your institution's name and marketing information. You can even print your graphic logo, if the printer is capable.

PAN Truncation

You can determine if the account numbers on a receipt are printed in full or are truncated so only some digits of the number appear. This option allows you to reduce the amount of fraud caused by carelessly discarded receipts falling into the wrong hands.

Transaction Surcharge

An optional surcharge can be added to certain transactions on your terminals. This surcharge is in addition to any fees that the cardholder's institution imposes; in other words, surcharges are for transaction acquirers, not issuers. This option is available only for Diebold 910 and 911 terminals.

You determine whether the surcharge is added, and the amount added, depending on such variables as whether the transaction is on-us or not-on-us, the account type of the transaction, and the card type. Terminals can also be grouped for surcharging purposes, for example, all of your terminals in airports can have the same surcharge amount. You also determine whether a surcharge is assessed when a partial amount is dispensed and the rest of the transaction is reversed.

Surcharging is customized through the Terminal Data File (TDF), which is documented in the ***BASE24-atm Files Maintenance Manual***, and the Surcharge File (SURF), which is documented in the ***BASE24 Base Files Maintenance Manual***. The surcharge amount is carried in the Surcharge Data token which is appended to the Standard Internal Message (STM). Tokens are described in the ***BASE24-atm Tokens Manual***.

The following cardholder transactions can have a surcharge added by BASE24-atm:

- Withdrawal from checking account
- Withdrawal from savings account
- Fast cash withdrawal
- Cash advance from credit account

Administrative Transactions

Administrative transactions can be performed at the terminal using an administrative card or they can be performed using the DCT. The administrative transactions available through BASE24-atm include the following transactions:

- Balance with current cash
- Balance with standard replenishment
- Terminal subtotals

- Increase cash
- Decrease cash

These terminal balancing transactions allow terminals to be balanced with either the standard replenishment balancing method or the current cash balancing method. The standard replenishment balancing method is typically used for Diebold 911 and 912 terminals which use currency cassettes. The current cash balancing method is used for Diebold 910 terminals which have open currency hoppers.

Diebold 912 ATMs can have up to four currency cassettes. Terminology for administrative transactions with the Diebold 912 terminals references cassette numbers 0 through 3 rather than hoppers with the highest or lowest bill denominations when administratively balancing cash in an ATM.

Balance with current cash is a terminal subtotals transaction which enables activity totals recorded since the last balancing transaction up to the present to be printed. Terminal totals are not cleared with this transaction.

Balance with standard replenishment is a midpoint cash adjustment transaction which allows terminals to be replenished by increasing or decreasing the bill denominations without balancing the terminal.

For more information on administrative transactions, refer to the ***BASE24-atm Settlement and Reporting Manual***.

Device Configuration and Control

Before a terminal is ready to accept transactions, it must be downloaded with configuration information. Configuration information is contained in the CNFG910 or the CNFG912. The information in this file is downloaded from a CRT controlled by the Device Control Terminal (DCT) functions. The CNFG910 is used with Diebold 910 and 911 terminals. The CNFG912 is used with Diebold 912 terminals. Additional information about the procedures used when downloading the CNFG910 or CNFG912 is available in the ***BASE24 Device Control Manual***.

The Configuration File is built and maintained through the use of the Diebold entry program. This program allows the states, screens, and timers in the CNFG910 or CNFG912 to be modified. The entry program is discussed in section 4 of this manual.

BASE24 allows terminals to be configured in a variety of ways by using different load groups. Load groups are listed and briefly described in section 2 of this manual.

Terminal Balancing

A terminal must be balanced on a regular basis once it is operational. This can be accomplished using BASE24 DCT functions or by initiating administrative transactions at the ATM. However, the centrally located DCT allows a terminal to be closed, balanced, and opened again without personnel visiting the terminal site. Other functions such as adding to the cash supply and physically servicing the terminal require administrative transactions which are performed at the terminal site.

Log Messages

The BASE24-atm Device Handler process generates log messages periodically and as needed to notify you of the state of an ATM. The messages range in severity from informational to critical and can be printed or displayed on BASE24-atm logs. Further information on log message generation is included in section 3.

Terminal Control

BASE24-atm provides several ways for you to control Diebold 910, 911, and 912 terminals. One of these ways is via the DCT. The DCT allows you to perform all of the functions necessary to control a terminal including starting and stopping, downloading, balancing, and inquiring into the status of the terminal.

The DCT is at the heart of the control mechanism for terminals within BASE24-atm. The DCT is the one module that contains all of the functions necessary to control the terminal. Anyone responsible for controlling terminals should have access to the DCT.

Other ways to impact terminals within BASE24 include issuing commands via a network control facility. Refer to the ***BASE24 Text Command Reference Manual*** for more information about network control facilities. Refer to the ***BASE24 CRT Access Manual*** and the ***BASE24 Device Control Manual*** for more information on how to access and use the DCT.

Basic Device Control Functions

The basic device control functions common to all types of terminals include the following:

- Balancing terminals
- Inquiring into the status of terminals
- Opening and closing the terminals
- Starting and stopping the terminals
- Displaying subtotals
- Increasing and decreasing cash totals in the terminal hoppers

Each of the functions listed above is associated with the DCT, since each is critical to the operation of an ATM. The centralized DCT allows these functions to be performed more efficiently. For example, when cardholders need to balance their terminals, they can close, balance, and reopen an ATM from the central DCT location using the common control facilities of the DCT.

BASE24-atm processes also can be reinitialized or warmbooted from the DCT. This is important because Device Handler processes must be reinitialized when changes that affect the Device Handler processes are made to the Logical Network Configuration File (LCONF). Refer to the ***BASE24 Device Control Manual*** for more information about device control functions.

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Section 2

ATM Configuration

BASE24 support of Diebold 910, 911, and 912 terminals includes flexible configuration options. There are numerous ways Diebold 910, 911, and 912 terminals can be configured, depending upon your preferences. For example, you could allow some terminals to offer certain transactions, while terminals in other locations would not allow the same transactions. You also can change the wording on their screens.

This section contains information about the following topics:

- The Diebold Configuration Files
- BASE24 files used to configure processing by Diebold ATMs

Diebold 910, 911, and 912 ATM Configurations

The Diebold 910, 911, and 912 ATMs are configured using software contained in the ATMs themselves. The Diebold internal software contains the control logic and associated parameters that determine how transaction processing is configured at the ATM level (for example, screen data and timer data).

In addition to the Diebold internal software, a number of BASE24 files are available to configure transaction processing at the network level to support the Diebold ATMs. The Diebold 910/911 and 912 Configuration Files (CNFG910 or CNFG912) are directly accessed by the ATM. BASE24 files provide indirect support to the ATM by defining the interaction between the Device Handler process and the ATM.

Because of the many combinations of options that are available during configuration, the Configuration Files and BASE24 files are discussed in this section one at a time. The various combinations that can be derived from these individual options are left up to you. The following files are used by the Diebold 910, 911, and 912 ATMs:

- Diebold Configuration Files (CNFG910 and CNFG912)
- Logical Network Configuration File (LCONF)
- Terminal Data File (TDF)
- Receipt File (RCPT)
- Network Environment Source File (NEFS)
- Institution Definition File (IDF)
- Log Message Configuration File (LMCF)

CNFG910 and CNFG912 Configuration Files

Changes can be made to the configuration files on the BASE24-atm system using the Diebold entry program and downloaded to the ATM from the BASE24 Device Control Terminal (DCT). Refer to the ***BASE24 Device Control Manual*** for information on the procedures used when downloading files. For further information on how to modify the configuration files, refer to section 4.

The CNFG910 and CNFG912 contain all of the state tables, screen definitions, financial institution tables (FITs), and timer information. State tables determine the screens that are displayed. Screen definitions determine what appears on the screen. The FIT contains control words for processing the PIN. Timer information in the CNFG910 and CNFG912 control ten timers, some of which control ATM time-outs and some that control message time-outs.

All of this information must be downloaded to the ATM, with the exception of the FIT if not applicable. This information is conveniently separated into groups; then the group can be downloaded to the ATM with all of the pertinent information being sent at one time. Refer to section 4 for a complete list of groups included in the CNFG910 and CNFG912. Groups 300 through 303 and 311 through 313 are described in section 5 of this manual.

One other group, besides Group 0 state tables, can be downloaded to the ATM if the appropriate field in the Terminal Data File (TDF) is set. This field is discussed in detail later in this section. The second group which is downloaded to the ATM overlays any duplicate screens in Group 0.

Although there are a number of different load groups, only two BASE24 groups can be downloaded to the ATM. However, since you have the ability to add additional state tables to either of the two groups, all the available state tables can be downloaded to the ATM if the CNFG910 or CNFG912 is modified appropriately.

You should note that each ATM can have a separate set of groups downloaded to it, depending upon the DCT command and the settings of the TDF. The TDF is discussed later in this section.

Logical Network Configuration File

The Logical Network Configuration File (LCONF) includes the locations for all the necessary files associated with a logical network. This includes the location of the CNFG910 and CNFG912, the Terminal Data File (TDF), and the Institution Data File (IDF).

The LCONF contains the records that link physical file names to the file names as they are referenced in BASE24 code and the records that set processing values for a logical network.

The records that link physical file names to the file names as they are referenced in code are known as assigns. The records that set processing values are known as params.

The LCONF assigns and params used by Diebold devices are listed below, including a short description of each. Refer to the ***BASE24 Logical Network Configuration File Manual*** for more detailed information on these LCONF assigns and params.

Assigns

The following assigns are used by the Diebold Device Handler processes to locate the appropriate files:

ATM-DH-CNF910 — The fully qualified file name of the Configuration File for Diebold 910 and 911 ATMs (for example, \B24.\$DATA.PRO1DATA.CNFG910). This assign is used by Diebold 910/911 Device Handler processes when downloading to terminals. This value can be overridden by a corresponding value in the device-dependent information in the individual TDF record for the ATM.

ATM-DH-CNF912 — The fully qualified file name of the Configuration File for Diebold 912 ATMs (for example, \B24.\$DATA.PRO1DATA.CNFG912). This assign is used by the Diebold 912 Device Handler processes when downloading to terminals. This value can be overridden by a corresponding value in the device-dependent information in the individual TDF record for the ATM.

LMCF — The physical file name of the Log Message Configuration File (LMCF). This file is used during the initialization and reinitialization of the Device Handler process.

ATM-RCPT — The physical file name of the Receipt File (RCPT). This file is used during the initialization and reinitialization of the Device Handler process.

TDF — The physical file name of the Terminal Data File (TDF).

IDF — The physical file name of the Institution Definition File (IDF).

Params

The following params are used for setting processing values at the logical network level:

ATM-DH-AUTH-TIMER — Specifies the number of seconds the Device Handler process waits for a reply from the BASE24-atm Authorization process before denying a request.

ATM-DH-MAX-TDFS — Specifies the amount of space to allocate for a table of TDF locations. This param represents the maximum number of TDF records associated with a single Device Handler process.

ATM-DH-MAX-IN-CORE — Specifies the maximum amount of extended memory to allocate for TDF records locked on disk for transactions in progress.

ATM-DH-OPT-LVL — Specifies the number of times the BASE24 Device Handler process writes to the TDF for normal transactions.

ATM-DH-STALE-RQST-TIMEOUT — Specifies the number of seconds an incoming transaction can remain in the queue of the Device Handler process before being discarded by the Device Handler process as too old to process.

SECURE-DEV-FAIL-NOTIFY — Specifies the interval that messages are logged due to transactions that are directed to an unavailable security device. Transactions directed to the primary security device are redirected to the secondary security device. Transactions directed to the secondary security device are declined if the device is unavailable. This parameter is accessed by security utility procedures during the initialization or reinitialization of the Device Handler process.

SECURE-DEV-TIM-LMT — Specifies the number of seconds to wait for a response from a security device. This parameter is accessed by security utility procedures during the initialization or reinitialization of the Device Handler process.

SECURE-DEV-TYP — Specifies the type of security module that is being used by the logical network. This parameter is accessed by security utility procedures during the initialization or reinitialization of the Device Handler process.

REVERSE-BAL-INQ — Specifies whether balance inquiry transactions can be reversed.

ATM-DH-ADDENDS — Contains a number used to display a set of screens in a secondary language.

DATE-DISPLAY-TYPE — A code that indicates the format of the month and day portion of the dates on reports and BASE24-atm Device Handler process receipts (that is, DDMM or MMDD).

ATM-DH-CONS-MAC-ERR-LMT — A code that indicates the number of consecutive message authentication code (MAC) errors that can occur before the Device Handler process sends a shutdown message to the ATM.

Terminal Data File

The Terminal Data File (TDF) record for an ATM defines much of the ATM functionality. Whenever BASE24 receives a message from an ATM, or a message is sent to an ATM, the TDF record for the terminal is accessed. A separate TDF record must exist for each ATM controlled by BASE24-atm.

A TDF record is accessed through nine screens using BASE24-atm file maintenance functions. The first six screens contain general information common to all ATMs; the last three screens contain information specific to Diebold ATMs. For complete information on the values of the individual fields in the TDF and how the TDF can be maintained, refer to the ***BASE24-atm Files Maintenance Manual***.

General TDF Information

The basic information contained in every TDF record includes the following:

- Reference information
- Transaction processing information
- Status information
- Transaction security information

Reference information includes information such as the physical location of the terminal, the owner of the terminal, and the Device Handler and Authorization processes involved in the transaction processing for the specified ATM.

Transaction processing information includes information on the types of not-on-us transactions allowed at the specific terminal. The Authorization process determines if a match exists between the sharing groups indicated in the TDF record for the specific ATM and the sharing groups set in the Institution Definition File (IDF) for the card issuing institution. This check determines the type of transaction agreement, if any, that has been made between the ATM owner and the institution owning the card and applies the terms of the agreement toward processing the cardholder's request.

Transaction message activity is monitored through a series of timer settings established in the TDF for peak and nonpeak periods and holiday schedules. Receipt groups for configurable receipts are also defined in the TDF.

Information on status changes to the ATM such as the state of the ATM and the amount of transaction activity for the ATM is also kept in the TDF.

You should note that several of the fields in the TDF are dependent upon the settings in the IDF. For example, the sharing arrangements between terminals and institutions, or in other words which institution's cardholders can initiate transactions at which terminals, are determined by comparing one field in the TDF with another field in the IDF. The field in the TDF is named GROUPS and is found on TDF file maintenance screen 1 and the field in the IDF is named SHARING GROUP and is found on IDF file maintenance screen 2. In order for cardholders from Bank B to initiate transactions at a particular terminal operated by Bank C, at least one of the characters in each of these fields must match.

Transaction security information relating to the encryption and decryption of data, local PIN verification, and message authentication through the message authentication code (MAC) is defined in the TDF.

Specific Diebold Information in the TDF

Diebold-specific file maintenance screens contain information such as the type of line protocol used with the terminal, the number of timers, and fast cash amount options. Some of the fields are dependent upon the hardware used. For example, one field includes information on the type of depository used; another indicates if the ATM has an audit printer. Other fields are dependent upon the preferences of the institution operating the ATM. For example, one set of fields determines whether receipts are mandatory, optional, or not generated for each allowable transaction.

Two fields are especially important for configuration issues. The first of these is the LOAD FILENAME field on TDF Diebold screen 7. The fully expanded name of the Configuration File, CNFG910 or CNFG912 file, should be placed in this field. If the name is not placed in this field, the Device Handler process uses the CNFG910 or CNFG912 file name found in the LCONF when the Device Handler process was last initialized.

The second field to be aware of is the TERMINAL GROUP field on TDF Diebold screen 7. This field is used to download a second group of state tables and screens from the CNFG910 or CNFG912. The Device Handler process checks this field when a download command is issued from the DCT to determine if another group besides Group 0 should be downloaded to the terminal. (Group 0 is always downloaded to the terminal on a full load.)

Transaction activity can be monitored through a series of timer settings established in the TDF for peak and nonpeak periods and holiday schedules. Receipt groups for configurable receipts are defined in the TDF.

Network Environment Source File

The Network Environment Source File (NEFS) is the source file used to create the Network Environment File (NEF). The NEF contains the parameters for controlling the environment within an XPNET node, including lines, stations, processes, and devices. Each line, station, process, and device in the XPNET node must be defined in the NEF in order to be recognized and used by BASE24. Therefore, every Diebold ATM in an XPNET node must have an entry in the NEF for the station and the line associated with the terminal.

Detailed information about how to add Diebold 910/911 and 912 declarations to the NEFS is included in the ***XPNET BASE24 Configuration Guidelines***.

Institution Definition File

The Institution Definition File (IDF) is used to define the processing parameters for each institution participating in the logical network. The IDF is read by the Device Handler process during an administrative transaction in order to retrieve the dates the Device Handler process requires for cutting over its terminals. In addition, the Device Handler process reads the holiday table in the IDF if terminals are configured, through the Terminal Data File (TDF), to support inactivity timers. Holiday dates are placed in the extended memory table. The IDF also contains routing tables used for transaction routing within BASE24, date parameters, transaction sets, and ATM processing control parameters.

For more information on the IDF, refer to the ***BASE24 Base Files Maintenance Manual***.

Receipt File

The Receipt File (RCPT) is used to support the configurable receipt option.

The BASE24-atm configurable receipt option provides you with the ability to establish ATM receipt formats that are unique to the institution. In addition, this customization option supports the display of an informational or marketing message, the use of flexible monetary symbols, multiple language support, and the option to print the account description and account type on the receipt. The configurable receipt option includes approved and denied financial transaction receipts as well as administrative receipts.

Receipt configuration is supported by receipt group and transaction type. Receipt groups allow the use of a single receipt format for a group of ATM terminals rather than configuring receipts for individual terminals. Receipt groups are specified for each terminal using screen 8 of the Terminal Data File (TDF). For further information on establishing receipt groups in the TDF and maintaining the RCPT, refer to the ***BASE24-atm Files Maintenance Manual***.

The RCPT is read by the Device Handler process to access the templates that will be used for configurable receipts. For a description of how the Device Handler process uses the RCPT in determining receipt verbiage and format as well as examples of default receipt templates for various transactions, refer to section 5.

Log Message Configuration File

The Log Message Configuration File (LMCF) contains one record for each log message that you have altered. Because LMCF records must be read into memory by the processes that use them, updates to the LMCF do not take effect until those processes are reinitialized.

An LMCF record is accessed using BASE24 operator functions. The file itself contains changes made to log messages in the form of records. Each log message routing code, severity, and text can be modified using BASE24 operator functions. Refer to the ***XPNET BASE24 Configuration Guidelines*** for more information about BASE24 operator functions and how to access the LMCF.

Section 3

BASE24-atm Device Handler Process

Device Handler processes are the primary interfaces between ATMs and BASE24. All messages that are sent by BASE24 to an ATM must first be received by the ATM's Device Handler process and then forwarded to the designated ATM. Likewise, all messages that are sent by an ATM to BASE24 must first be received by the ATM's Device Handler process.

Because each type of ATM has its own message format and way of processing transactions, separate Device Handler processes must exist for each type of terminal supported by BASE24. For example, Diebold 910 and 911 ATMs have their own type of Device Handler process, which differs from the Device Handler process used for Diebold 912 and 1000-series, NCR 50XX/56XX-series, IBM or other ATM.

This section provides the following:

- An overview of the BASE24-atm 910/911 and 912 Device Handler processes.
- Descriptions of processing steps taken by the Device Handler process when it is first started.
- A list of BASE24 commands received by the Device Handler process to perform specialized types of activities.
- Information on how to access the log messages generated by the Diebold 910/911 and 912 Device Handler processes.

Diebold 910/911 and 912 Device Handler Processes

Diebold 910, 911, and 912 terminals are linked to BASE24-atm processing by a Device Handler process tailored for Diebold 910, 911, or 912 terminals. Every transaction initiated at a Diebold 910, 911, or 912 terminal is passed to the Device Handler process specified for 910/911 or 912. The message coming from the terminal is in the native mode of the device, and the Device Handler process converts it into the BASE24 Standard Internal Message (STM).

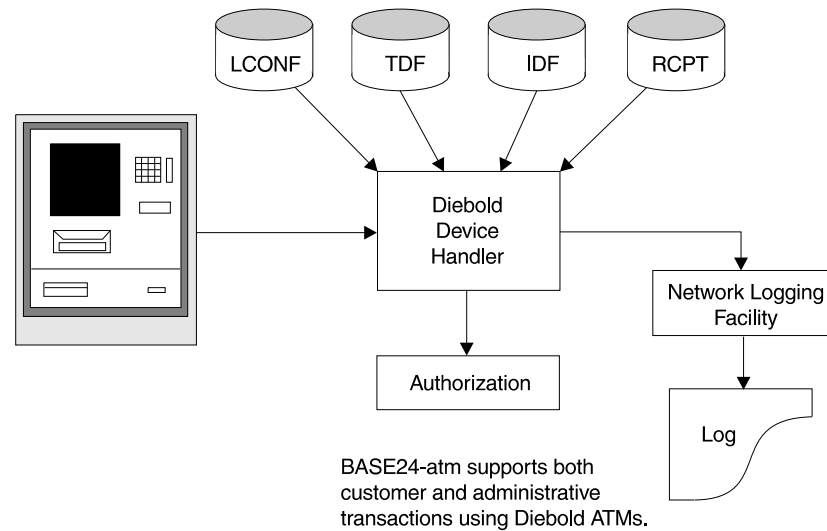
Transactions processed at different Diebold 910, 911, or 912 terminals can be handled differently depending upon how the individual terminal is set up in the Terminal Data File (TDF). This allows each terminal to have different sharing arrangements, available transactions, and PIN encryption methods.

The Device Handler process reads the TDF whenever it needs information about a terminal. The Device Handler process obtains the name of the appropriate TDF by reading the Logical Network Configuration File (LCONF).

The Institution Definition File (IDF) is another file read by the Device Handler process. The IDF is used to obtain the dates the Device Handler process requires for cutting over its terminals. It is also used to determine the holidays established by an institution for use in supporting the inactivity timer option. Inactivity timers enable an institution to monitor transaction activity during specified periods.

The Receipt File (RCPT) is read by the Device Handler process to determine the verbiage and format of financial and administrative receipts.

When problems occur with the terminal, the Device Handler process notifies BASE24's network logging facility. The network logging facility then writes an appropriate error message to as many as three printers or terminals. The following graphic shows the role of the Device Handler process during transaction processing.



Initialization Steps

The following steps are taken by the Device Handler process when it is first started or initialized:

1. Initializes its timer table.

When a Device Handler process creates timers, it places them in its timer table. Device Handler processes maintain three types of timers:

- The Authorization Response timer times each authorization response. If an authorization response is not returned to the Device Handler process within this time limit, the request is automatically denied. This timer is set via the ATM-DH-AUTH-TIMER param in the LCONF.
- The Download 910/911 and 912 timer (only kept by Device Handler processes that are capable of doing downloading) times each station download record. If no response is received from the station within this time limit, the Device Handler process automatically aborts the download. This timer is always set to 45 seconds.
- The Inactivity timer indicates a period of time during which some activity should occur at the terminal. If no activity is determined, the Device Handler process logs a message. This timer is set on TDF screen 7, with different settings for workday and weekend/holiday peak usage time periods. This timer allows a maximum number of inactivity timeouts according to the maximum, which is also specified on TDF screen 7. Once the maximum is reached, terminal monitoring is discontinued.

2. Reads the name of the LCONF into memory.

The startup message from the XPNET process contains the name of the LCONF, which the Device Handler process reads into memory. This name allows the Device Handler process to access the LCONF for the assigns and params it uses in processing.

3. Reads the following assigns from the LCONF:

IDF
TDF
LMCF
ATM-RCPT
ATM-DH-CNF910 or ATM-DH-CNF912

4. Reads the LMCF for its modified log messages.

The LMCF contains a record for each modified log message in the system. The Device Handler process opens the LMCF, retrieves its own modified log messages, and closes the file.

5. Initializes the network logging facility for writing log messages.
6. Loads all params shown in the following param list.

ATM-DH-ADDENDS
ATM-DH-AUTH-TIMER
ATM-DH-MAX-TDFS
ATM-DH-MAX-IN-CORE
ATM-DH-OPT-LVL
ATM-DH-STALE-RQST-TIMEOUT
REVERSE-BAL-INQ
DATE-DISPLAY-TYPE
ATM-DH-CONS-MAC-ERR-LMT

- a. Reads and edits the ATM-DH-ADDENDS param in the LCONF.

If the ATM-DH-ADDENDS param is invalid or cannot be found, the Device Handler process uses a default value that is normally an appropriate value for the specific type of ATM.

- b. Reads and edits the ATM-DH-MAX-IN-CORE and ATM-DH-MAX-TDFS params in the LCONF.

If the ATM-DH-MAX-IN-CORE param or the ATM-DH-MAX-TDFS param is invalid or cannot be found, the Device Handler process uses default values.

- c. Reads and edits the ATM-DH-OPT-LVL param in the LCONF.
- d. Reads and edits the DATE-DISPLAY-TYPE param in the LCONF.
- e. Reads and edits the ATM-DH-CONS-MAC-ERR-LMT param in the LCONF.

If the ATM-DH-CONS-MAC-ERR-LMT param is invalid or cannot be found, the Device Handler process uses a default value of 3.

- f. Reads and edits the ATM-DH-AUTH-TIMER param in the LCONF.
- g. Reads and edits the ATM-DH-STALE-RQST-TIMEOUT PARAM in the LCONF.
- h. Reads and edits the REVERSE-BAL-INQ param in the LCONF.

7. Opens the TDF with read/write access using the name specified in the TDF assign in the LCONF.

8. Opens the RCPT with read access using the name specified in the ATM-RCPT assign in the LCONF.
9. Initializes an extended memory area for finding and keeping TDF records.

The Device Handler process uses the ATM-DH-MAX-TDFS param in the LCONF to build a table of the TDF's key record positions.

The Device Handler process uses the ATM-DH-MAX-IN-CORE param in the LCONF to determine how much memory is needed to keep copies of the TDF records in extended memory. This param indicates the maximum number of transactions that can be processed by a Device Handler process at one time.
10. Initializes RCPT by reading all RCPT records with device type of "D910" or "D912" into extended memory. Closes receipt file.
11. Opens the IDF with read access using the name specified in the IDF assign in the LCONF.
12. Calls a security utility procedure that attempts to open the security device, if a security device is being used.

The security devices that are used by the network are specified in the SECURE-DEV1 and SECURE-DEV2 assigns in the LCONF.

If the security devices cannot be opened, a log message is created and sent to the logging facility indicating that one or both of the security devices could not be opened.

The SECURE-DEV-FAIL-NOTIFY, SECURE-DEV-TIM-LMT, and SECURE-DEV-TYP params are also accessed in the LCONF at this time by the security utility procedure.
13. Initializes PAMS (if bound in).
14. Reads into extended memory a table of holidays to be used with the inactivity timers.
15. Closes the LCONF.

Warmboot Procedures

The warmboot command (9501****) is a general use operator command sent using the Device Control Terminal (DCT) or a network control facility. General use operator commands enable an operator to control or change the way the Device Handler process is operating.

When Device Handler processes are warmbooted, they do not go through their entire initialization routine. Instead, they only perform the following steps:

1. Reread the LCONF.
2. Close and reopen the files they use.
3. Release the current extended memory.
4. Close and reopen the security module.
5. Reinitialize the LMCF.
6. Reallocate new extended memory based on the current settings in the LCONF.
7. Reinitialize the TDF extended memory data, RCPT extended memory data, and IDF holiday extended memory tables.
8. Reinitialize PAMs module (if PAMs is being used).

For detailed information on the warmboot command and other operator commands, refer to the ***BASE24 Text Command Reference Manual***. Refer to the ***BASE24 Device Control Manual*** for information on the DCT.

Device Handler Text Commands

Text commands are used to send information to a process. You can enter these commands by using a network control facility. These commands serve as a tool that you can use to perform network management functions such as changing certain processing parameters, warmbooting a specific process, monitoring a specific process, and resolving problems encountered by a specific process. The following text commands can be used with the BASE24-atm Diebold 910/911 and 912 Device Handler processes:

- The OPEN command causes the Device Handler process to send the appropriate native-mode message to open the ATM.
- The CLOSE command causes the Device Handler process to send the appropriate native-mode message to close the ATM.
- The LOAD commands direct the Device Handler process to download some form of configuration information to the ATM.
- The RESET command causes the Device Handler process to interrupt any current transaction or load in progress for the given ATM. You should not enter this command unless requested to do so by ACI personnel.
- The warmboot command (9501****) directs the Device Handler process to warmboot. For a list of the steps that the Diebold 910/911 or 912 Device Handler processes follow when a warmboot command is issued, refer to the topic “Warmboot Procedures” above.

For detailed descriptions and the syntax of all BASE24 text commands, refer to the ***BASE24 Text Command Reference Manual***.

Log Message Generation

The Diebold 910/911 and 912 Device Handler processes generate log messages that keep operators aware of activity occurring in the system. Log messages can be used to assist operators with their daily tasks or to notify operators of problems occurring in the system. Log messages provide operators with insight as to how the system is running.

For a complete explanation of log messages and for information on how to access the documentation for log messages generated by the Diebold 910/911 and 912 Device Handler processes, refer to the ***BASE24 Log Message Reference Manual***.

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Section 4

Configuration File Maintenance

The Diebold 910/911 and 912 Configuration Files (CNFG910 and CNFG912), which are resident in the BASE24-atm system, can be accessed using the Diebold entry program and downloaded using the Device Control Terminal (DCT).

With access to the Configuration File, you can make changes to ATM screen displays, state table entries, FIT data, configuration parameters, and timers. These changes can then be downloaded to the ATM.

This section details the BASE24-atm screens you can use to make changes to the Configuration File using the Diebold entry program.

Information on downloading Configuration File information to Diebold 910, 911, and 912 ATMs can be found in the ***BASE24 Device Control Manual***.

Local PIN Verification

BASE24-atm supports local PIN verification in which Diebold 910, 911, and 912 terminals, which have PIN verification capability, perform PIN verification at the device for financial and administrative transactions without the involvement of the BASE24 network. For complete information on local PIN verification, refer to the ***Diebold Data Security Procedures and Reference Manual, October, 1983***.

Local PIN verification requires changes to the financial institution table (FIT), the PIN Entry state, and the Transaction Request state in the CNFG910 and CNFG912.

The following changes must be made to the PIN Entry state, the Transaction Request state, and the FIT to support local PIN verification. The configuration screens used to make these changes are described later in this section.

PIN Entry State

The GOOD PIN NEXT STATE field in the PIN Entry state must be configured to specify the next state to which the terminal proceeds upon successful local PIN verification. The PIN MODE/NEXT STATE NO PIN CHK field in the PIN Entry state must contain the value 255 (not used).

Transaction Request State

The SEND BUFFER DATA in the Transaction Request state must be set to 000 (do not send PIN buffer).

FIT parameters must be modified to support local PIN verification. Refer to the appropriate Diebold vendor documentation for complete information on setting up the FIT data to support local PIN verification.

Financial Institution Table

The FIT data specifies parameters to be used depending on the terminal PIN verification process to be used. The following field values are important to support local PIN verification.

Field	Value	
PINPD	Must be set to 000.	
PRCNT	The value is 255 indicating the PIN-retry count is not on the ATM card, but in the PIN Entry State (B) table.	
PDCTB	The institution is responsible for the values for this field.	
PEKEY	The institution is responsible for the values for this field.	
PINDX	The fifth PINDX digit is reserved and must be hexadecimal 0.	
PCKLN	The valid values for the entire PCKLN byte, excluding Identkey, are shown below.	
	Terminal Verification	PCKLN ByteValue
	* None	000
	DES	004–016
	Diebold	064–068
	VISA	032–036

* The terminal does not perform local PIN verification.

Diebold Entry Program

The Diebold ATMs are controlled by software that is stored in a configuration file on the BASE24 system and downloaded to the ATM. This configuration file is maintained on disk and is accessed through the Diebold entry program, also known as the Diebold Configurator process.

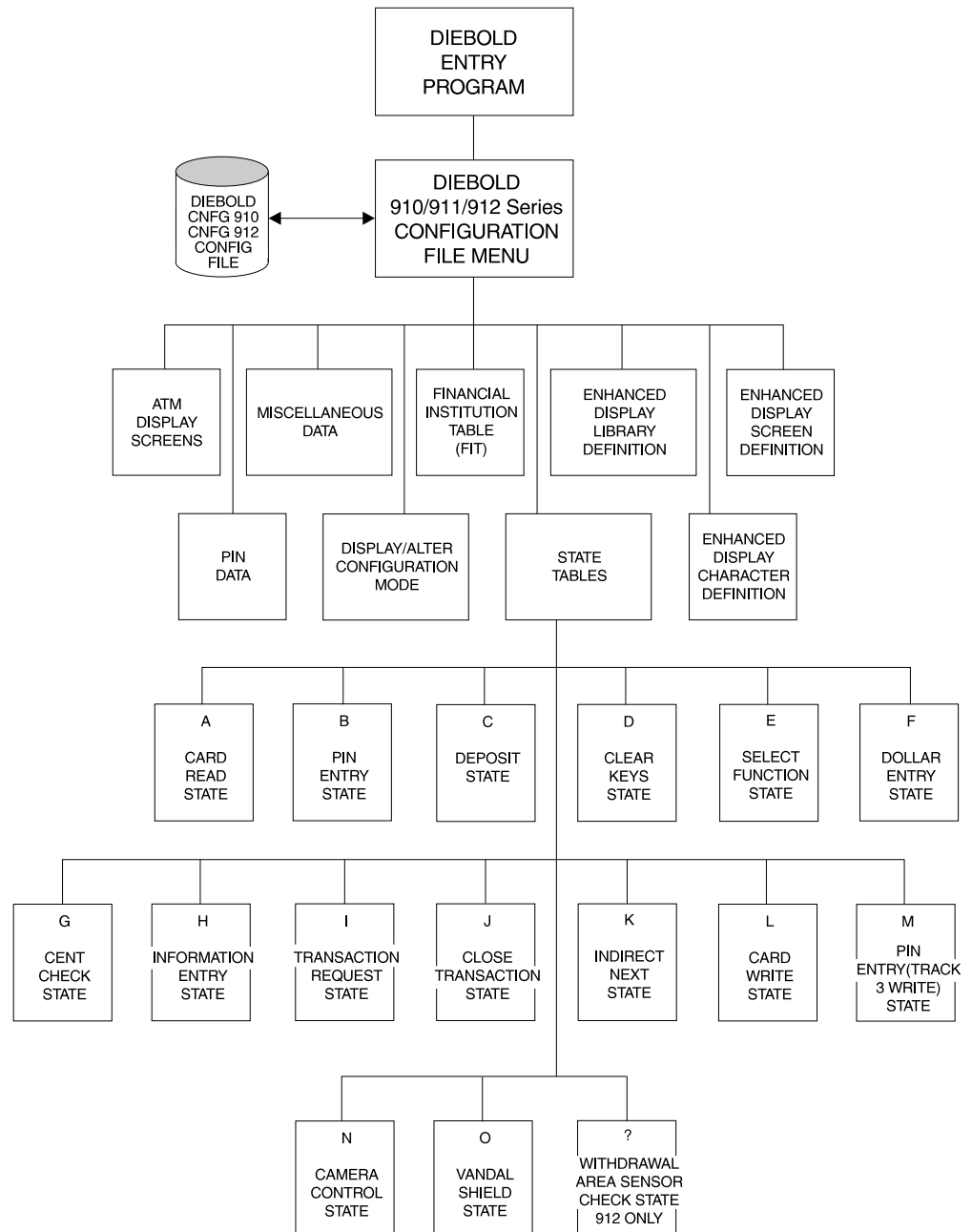
The Diebold entry program allows you to define the following attributes:

- ATM display screens
- State tables
- Miscellaneous data
- Financial institution table data
- Enhanced graphics

Additional information on configuration issues is found in section 2 of this manual. The remaining information in this section deals with the mechanics of running the Diebold entry program.

Diebold Entry Program Screen Map

The following screen map depicts the topology of the screens that are used to modify the CNFG910 or CNFG912 using the Diebold entry program.



Accessing the Diebold Entry Program

The Diebold entry program can be used to inquire, add, delete, change, and print the CNFG910 and CNFG912. You can specify the name of the Configurator process when you run the Diebold entry program. The process normally resides on subvolume AT51OBJ. (Refer to the Tandem ***Guardian Users Guide Manual*** for more information on the RUN command). To access the Diebold entry program and these functions, the following command must be entered from the Tandem TACL prompt.

Note: Before running the Diebold entry program, verify that the program and its corresponding library file are of the most recent version. If the file versions are not compatible, the Diebold entry program can run, but the results of any modifications cannot be guaranteed.

Syntax

RUN ENT9000/[IN *configfile*][,OUT *listfile*]/ [\$*terminal*]

IN *configfile* — Specifies the file name of the Configuration File to be accessed. This name must be entered using the Tandem file naming convention of \$*volume.subvolume.filename*. If omitted, the default is CNFG910 on the current volume and subvolume.

OUT *listfile* — Specifies the device where print functions direct their output. If omitted, the default is \$\$.#HOLD.

\$*terminal* — Specifies the terminal from which you perform the file maintenance. It must be a Tandem 6520/6530 terminal. If no terminal is specified, the default is the home terminal from which the program was started.

Example

RUN ENT9000/IN \$data.at51data.CNFGIX,OUT \$s.#cnfg/

Diebold Configurator Process Menu

When the Diebold entry program is run, the system displays the menu for the Diebold entry program, known as the Diebold Configurator Process menu. The menu is used to select the type of Configuration File data to be accessed, to print selected parts or all of the Configuration File, and to create a new Configuration File.

The Diebold Configurator Process menu is shown below, followed by field descriptions.

```

                                DIEBOLD CONFIGURATOR PROCESS - VERSION 2.24 (07Apr89)

Code  Functional Description
-----
  1.  ATM Display Screens    ***> PRINT functions from this menu generate cross-
  2.  State Tables          ***> references of Displays, Overlays, and States.
  3.  Miscellaneous Data
  4.  Pin Data
  5.  FIT (Financial Institution Table)
  6.  Display/Alter Configuration Mode
  7.  Enhanced Display Library Definition    ***> Enhanced Displays and Standard
  8.  Enhanced Display Character Definition ***> Displays are mutually-exclusive
  9.  Enhanced Display Screen Definition    ***> and depend on ATM's strapping.

Enter Function Code (from above menu) and press appropriate function key ...

Diebold Configuration File Name ..... $NOVA.AT40C910.CNFG910
Output Device for PRINT function ..... $S.#HOLD

=====
F1=ENTER      F3=CREATE FILE      F10=PRINT      F16=EXIT
                SHIFT-F10=PRINT FILE

```

Enter Function Code — A code identifying the configuration data to be modified. The list at the top of the screen identifies the valid values. The **F1** key must be pressed to enter the desired code. The system responds by displaying the appropriate screen for modifying the selected configuration data.

Valid values are as follows:

- 1 = ATM Display Screens. This code allows you to access the Diebold ATM high-level and low-level display screens.
- 2 = State Tables. This code allows you to access the Diebold State Table screen.
- 3 = Miscellaneous Data. This code allows you to access the Diebold Miscellaneous Data screen.
- 4 = Pin Data. This code allows you to access the Diebold PIN Data screen.

- 5 = FIT (Financial Institution Table). This code allows you to access the Diebold Financial Institution Table screen.
- 6 = Display/Alter Configuration Mode. This code allows you to access the Diebold Display/Alter Configuration Mode screen.
- 7 = Enhanced Display Library Definition. This code allows you to access the Diebold Enhanced Display Library Definition screen.
- 8 = Enhanced Display Character Definition. This code allows you to access the Enhanced Display Character Definition screen.
- 9 = Enhanced Display Screen Definition. This code allows you to access the Enhanced Display Screen Definition screen.

Note: Function codes 7, 8, and 9 are valid only for Diebold 912 terminals. If you choose not to proceed with screen 7, 8, or 9, any function key other than **F1** should be pressed to stop the process. If you enter a different screen number in this field and press the **F1** key, the first screen entered (7, 8, or 9) is accessed.

Field Length: 1 numeric character
Required Field: Yes, if the **F1** key is pressed
Default Value: No default value

Diebold Configuration File Name — The file name of the Diebold Configuration File. This name must be entered using the Tandem file naming convention of *\$volume.subvolume.filename*.

When the file name entered does not exist, the system displays a message at the bottom of the screen advising that the file name entered does not exist. You then select another file or create a new Configuration File.

To create a new Configuration File, enter the new file name directly over the displayed file name and press the **F3** key. The system responds by creating the new file and displays a message at the bottom of the screen indicating that the new file was created.

At times, the file can be opened with READ-ONLY access due to a possible security violation or because another process has the CNFG910 or CNFG912 open. The message "FILE CAN ONLY BE OPENED READ-ONLY" is displayed at the bottom of the screen. When this occurs, all functions can still be performed except for add, delete, or update.

If the file can only be opened with READ-ONLY access, you should check for a possible security violation. If another process has the file open, you must select another file or wait until the other user is finished.

Field Length: 35 alphanumeric characters
Required Field: Yes
Default Value: *\$currvol.currsubv.CNFG910*, where *currvol* is the current volume and *currsubv* is the current subvolume

Output Device for PRINT Function — The name of the output device for items to be printed.

This field is required and can be either a printer (or the Spooler), an entry-sequenced file, or another terminal. It cannot be the same terminal where the Diebold entry program is running.

When the output device named is another terminal, the Diebold entry program performs paging and prompts you to press the **Return** key before starting a new page. To stop the print function, press the **Control-Y** keys.

Whenever a print function is in progress, the terminal keyboard is locked (all keyboard functions are disabled). Once the print function is completed, the keyboard is unlocked.

Example: *\$\$.#PLP*
Field Length: 35 alphanumeric characters
Required Field: Yes, if the **F10** key is pressed
Default Value: *\$\$.#HOLD*

Function Keys

The use of one function key on the Diebold Configurator Process screen varies from the standard function keys explained in the ***BASE24 CRT Access Manual***. In addition, one key which was not assigned a function in the standard definitions is assigned a special function for this screen. The use of these function keys is explained below.

The first column of information below shows the BASE24 keys. The second column describes the functions that can be accomplished with these function keys.

Key	Description
F3	CREATE FILE — Creates a new Configuration File.
Shift-F10	PRINT FILE — Prints the contents of the Configuration File to the specified output device.

Configuration File Data

The following are Diebold entry program screens that are used to modify the Configuration File.

ATM Display Screens

ATM display screens relay information and ask for input from the cardholder at the ATM. Display screens can be modified using the entry program.

State Tables

The state tables screens allow you to configure the state tables. For example, you can configure the card read state, the PIN entry state, clear keys state, and dollar entry state. Refer to the topic “State Tables” for a complete list of states.

Miscellaneous/Timers

Three miscellaneous data fields and 10 internal time-out values may be altered using the entry program. The time-out values include cardholder response time, cardholder time-out response, close screen display time, depository response time, and delivery door time-out.

Pin Data Screen

The Pin Data screen includes PIN information if the terminal is not set to do DES PIN encryption.

Financial Institution Table Screen

The Financial Institution Table screen contains PIN verification information, institution IDs, routing information for multi-institution terminal use, and data on where to find specific information on a card.

Display/Alter Configuration Mode Screen

The Display/Alter Configuration Mode screen allows you to change the size of the display screen and the type of data shown in low-level mode.

Enhanced Display Library Definition Screen

The Enhanced Display Library Definition screen allows you to set the size of the character set(s) displayed on the ATM's CRT. If used, it replaces the ATM display screen function defined above. The ATM must contain the proper firmware to be able to use this feature. The Diebold 912 Device Handler process provides support for this feature.

Enhanced Display Character Definition Screen

The Enhanced Display Character Definition screen allows you to define nonalphabetic character displayed on the ATM's CRT. If used, it replaces the ATM display screen function defined above. The ATM must contain the proper firmware to be able to use this feature. The Diebold 912 Device Handler process provides support for this feature.

Enhanced Display Screen Definition Screen

The Enhanced Display Screen Definition screen allows you to design a screen to be displayed on the ATM's CRT that uses the character set(s) defined on the Enhanced Display Character Definition screen. If used, it replaces the ATM display screen function defined above. The ATM must contain the proper firmware to be able to use this feature. The Diebold 912 Device Handler process provides support for this feature.

Configuration File Function Keys

The function keys specific to most of the screens associated with the Diebold 910/911 and 912 Configuration Files are described below.

The first column of information below shows the BASE24 keys. The second column describes the functions that can be accomplished with these keys. The BASE24 functions activated by these keys can vary depending on which menu or screen they are associated.

Key	Description
F1	ENTER — Checks the data that has been entered on the screen for errors and enters the data in the file.
F2	READ DATA — Reads data in the file.
F3	ADD DATA — Adds data to the file. The data added must be unique within the file.
F4	DELETE DATA — Deletes data from the file.
F5	UPDATE DATA — Changes data already in the file.
F9	NEXT — Reads the next record. At the end of file, an error message is written at the bottom of the screen indicating end-of-file.
F10	PRINT SCREEN — Sends the screen currently being displayed to the spooler location specified on the Diebold Configurator Process screen in the Output Device For PRINT Function field as explained in this section.

Key	Description
F13	CONFIG SCREEN — Accesses the Display/Alter Configuration Mode screen. Although this function key is not listed on the bottom of the screen, it can be used from any screen.
F16	EXIT — Exits the Diebold entry program or the configuration screen currently being accessed. This function key is operable on the Help screen to enable you to resume normal processing.

ATM Standard Display Screens

The standard ATM display screens can be accessed by entering function code 1 on the Diebold Configurator Process screen. These screens allow modification of the screens displayed on the ATM.

ATM screens can be displayed in a low-level or in a high-level mode for changing screens. You can move between the two modes by pressing the **F1** key. The low-level mode formats serve little purpose for Diebold 910, 911, and 912 terminals other than to allow you to see the hexadecimal codes that correspond to the text on the display screen.

The Diebold standard display screens for high-level display and low-level display are described below and on the following pages.

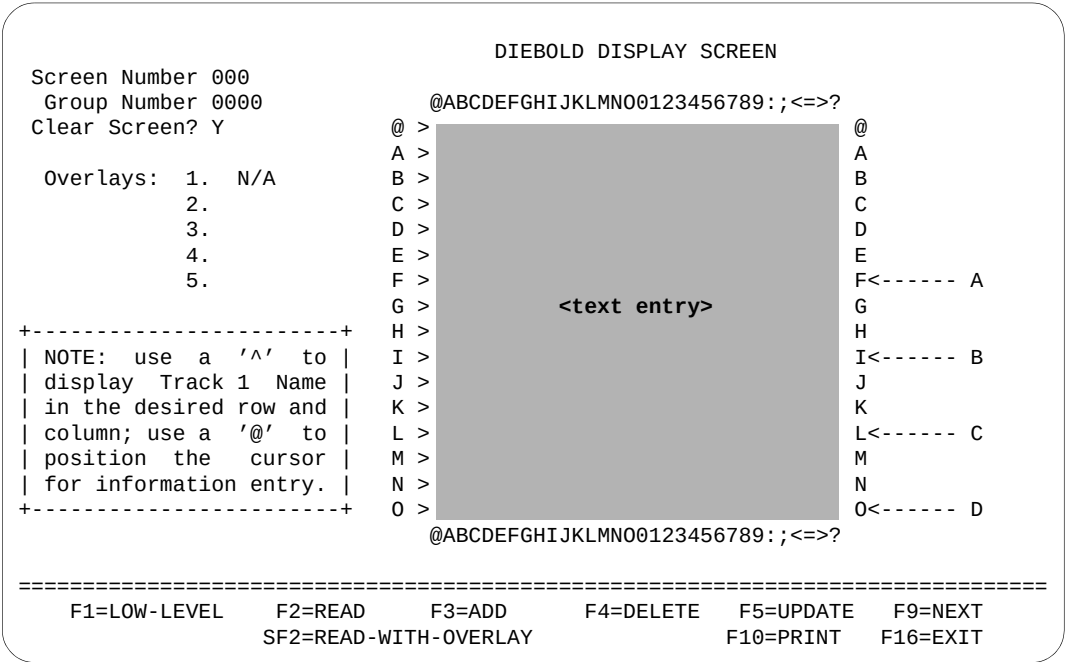
Diebold display screens 4 through 9 are restricted to certain formats; the Diebold entry program issues a warning before allowing access to these screens. Screens 0 through 3 also have a special meaning to the Diebold. The Diebold entry program issues a warning when attempting to ADD or UPDATE screens 0 through 3; each must contain a specific message:

Screen No.	Screen Message
Screen 0	The cardholder has taken too much time to make a selection.
Screen 1	The ATM is offline.
Screen 2	The ATM is out of service.
Screen 3	The ATM is being serviced.

If the complete screen table is printed (that is, the **F10** or **Shift-F10** key is pressed as shown on the menu), a cross-reference listing of overlayed screens is generated showing which screens are used as overlays for other screens.

Diebold High-Level Standard Display Screen

The High-Level Standard Display screen is shown below, followed by field descriptions.



Screen Number — A number between 000 and 255 that is associated with the desired screen.

Field Length: 3 numeric characters
Required Field: Yes
Default Value: 000

Group Number — The number of the group associated with the desired screen. The valid group numbers are different for 910/911 terminals than for 912 terminals.

ACI supplies sixteen groups of data in the CNFG910 for Diebold 910 and 911 terminals. These groups consist of the following:

- 0 = Basic set of states and screens for all ATM transactions; this group is always downloaded to the terminal on a full load. It is possible to load only the group specified in the TDF using the Device Control Terminal (DCT).
- 3 = Set of states used for whole-dollar amount entries.
- 5 = States and screens used for fast cash transactions.
- 7 = States and screens used for multiple languages support.
- 11 = States and screens used for Diebold 911 terminals without a withdrawal door.
- 20 = States and screens used for ANSI PIN block support.
- 100 = States and screens used for local PIN verify support.
- 200 = States and screens used for MAC Multiple Account Selection (MAS) support.
- 300 = The basic group for the optional receipts function. This group is merged with group 0.
- 301 = Optional receipts. The printer currently is operational. The cardholder selects optional receipts once for all chained transactions, on all approved or approved and denied transactions.
- 302 = Optional receipts. The printer currently is inoperable. The cardholder selects optional receipts once for all chained transactions.
- 303 = Optional receipts. The printer currently is operational. The cardholder selects optional receipts once for all chained transactions, only if the printer becomes inoperable.
- 304 = States and screens used for surcharging support.
- 311 = Optional receipts. The printer currently is operational. The cardholder selects optional receipts before each transaction, on all approved or approved and denied transactions.
- 312 = Optional receipts. The printer currently is inoperable. The cardholder selects optional receipts before each transaction.
- 313 = Optional receipts. The printer currently is operational. The cardholder selects optional receipts before each transaction, only if the printer becomes inoperable.

ACI supplies seven groups of data in the CNFG912 for Diebold 912. These groups consist of the following:

- 0 = Basic set of states and screens for all ATM transactions; this group is always downloaded to the terminal using the DCT. It is possible to load only the group specified in the TDF using the Device Control Terminal (DCT).
- 1 = Set of states and screens used for multiple languages support.
- 3 = Set of states used for whole-dollar amount entries.
- 5 = States and screens used for fast cash transactions.
- 14 = States and screens used for withdrawal area sensors support.
- 24 = States and screens used for encrypted PIN change support.
- 100 = States and screens used for local PIN verification support.

Field Length: 1–4 numeric characters

Required Field: Yes

Default Value: 000

Clear Screen — Specifies whether the ATM screen is cleared before this screen is displayed. Valid values are as follows:

Y = Yes, clear the ATM screen before displaying the current screen.

N = No, do not clear the ATM screen before displaying the current screen.

Field Length: 1 alphabetic character

Required Field: Yes

Default Value: Y

Overlays — Specifies a screen number or numbers to use as an overlay for this screen. Any screen referenced in this field should have its CLEAR SCREEN flag set to N. Up to five screens can be overlaid. Valid values are 000 through 255.

Field Length: 3 numeric characters

Occurs: Up to 5 times

Required Field: No

Default Value: No default value

<text entry> — This portion of the screen is used to create the ATM screen. There can be up to 40 characters of screen text on each of the 20 lines on the screen. The following two special characters, which are unique to the Diebold ATM, can be entered when using the High-Level Standard Display screen:

- @ = Indicates where to position the cursor on an information entry screen. The @ character does not display on the Diebold screen, but must be filled in during entry. The @ character must be entered on a line by itself. Only one @ character can be defined on a given screen. In low-level mode, this character shows up as a trailing “set cursor,” or the low-level code SI, *row.column*.
- ^ = Indicates where to display Track 1 data on the screen. (Track 1 usually contains the cardholder’s name.) It need not be the only character on a line, but must be the last character on the line. Multiple ^ characters can be defined on a screen, but the Diebold entry program issues a warning before allowing you to continue. In low-level mode, this character shows up as the low-level codes SI, *row.column*, HT.

Function Keys

The use of two function keys on the ATM standard display screens varies from the standard function keys explained earlier in this section. The use of these function keys is explained below.

Key	Description
F1	LOW LEVEL — Displays the low-level screen. When this function key is pressed from the high-level screen, the low-level screen is displayed. When this function key is pressed from the low-level screen, the high-level screen is displayed.
Shift-F2	READ-WITH-OVERLAY — Displays any overlays associated with the screen currently being accessed.

Warning Messages

The following warning messages are specific to the high-level screen and are generated when something that is not recommended has been done. For institution-specific reasons, however, certain programming practices may cause these warning messages to be displayed. You can override these messages by pressing the **F1** key.

Cursor Positioning is Normally Done by "Calling" Screen

Cause: The screen currently displayed is an overlay and is set to position the cursor.

Remedy: Screens that call overlays normally control cursor positioning. If the screen that calls this overlay is set up otherwise, the warning message can be overridden safely.

Typically a Screen Would Use the Same Overlay Only Once

Cause: The same overlay has been used more than once with the same screen. Unless the low-level mode has been used to set the cursor before calling each overlay, this is unnecessary since each occurrence will occupy the same space on the screen.

Remedy: Verify that a single overlay is needed multiple times. If it is not needed, remove the duplicate reference.

Typically a Screen Would not be Its Own Overlay

Cause: You have indicated that the screen currently displayed should overlay itself. Usually, this situation results in an infinite loop.

Remedy: Remove the use of this screen as its own overlay.

Note: Low Level Access is Recommended for Updates *reason*

Where: *reason* = Imbedded Cir Screen; No Trailing Set Cursor; Trailing Set Cursor; or Imbedded Overlays Used

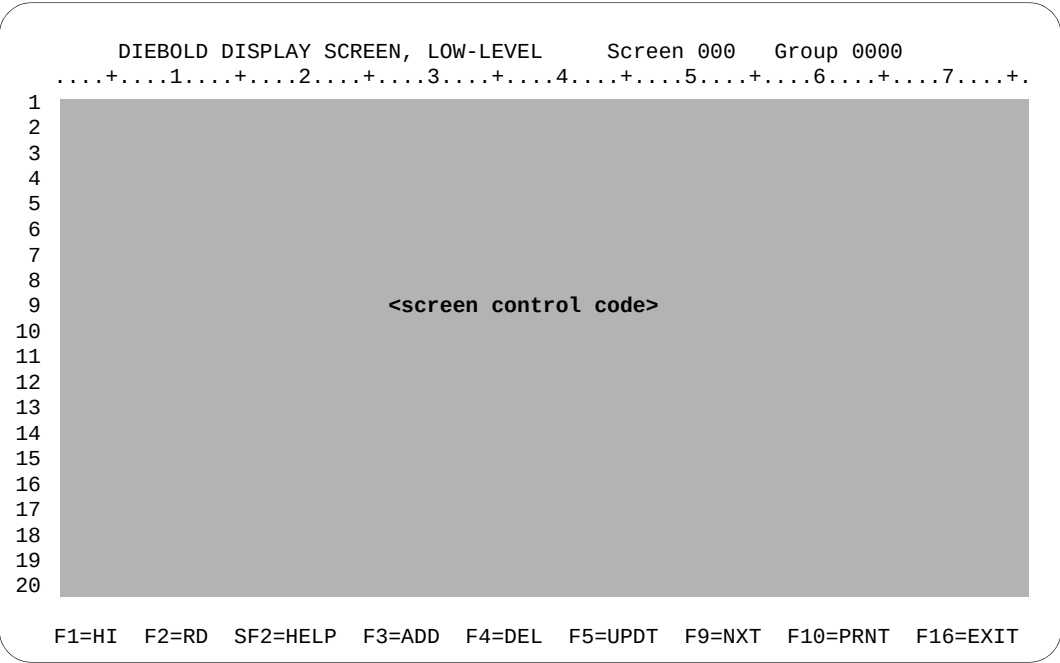
Cause: One of several warnings was overridden from the low-level mode and you are now attempting to read the screen associated with the override in the high-level mode.

Remedy: While it is possible to update screens from the high-level mode when this message is displayed, the screen display may be different than it appears. It is recommended that the screen be updated from the low-level mode when this message is displayed.

Diebold Low-Level Display Screen

When in the low-level mode, the data can be displayed in three different formats. To access these formats, you must change the settings on the Display/Alter Configuration Mode screen. The Display/Alter Configuration Mode screen can be accessed from this screen by pressing the **F13** key. If the settings on the Display/Alter Configuration Mode screen are changed, the changed display format will be reflected when a screen is read from the low-level screen.

The Diebold Display Screen, Low-Level is shown below, followed by field descriptions.



Screen — A number that is associated with the desired screen. Valid values are 000 through 255.

Field Length: 3 numeric characters
Required Field: Yes
Default Value: 000

Group — The group number used to identify a particular load group such as a collection of states, screens, etc., that is downloaded to the ATM. Each group provides certain functionality such as, Group 0 is the basic group, Group 5

provides fast cash support, and so on. The group number of each ATM is assigned in the TERMINAL GROUP field of the TDF. Refer to the above topic “High-Level Standard Display Screen” for a complete list of groups.

Field Length: 1–4 numeric characters
Required Field: Yes
Default Value: 000

<screen control code> — A variable number of codes that control how the screen is displayed to the cardholder. Place a comma between multiple codes on the same line. There can be up to 76 characters of codes on each of the 20 lines on the screen. A complete list of codes supported by BASE24-atm follows the description of this screen.

Function Keys

The use of two function keys on the low level screen varies from the standard function keys explained earlier in this section. The use of these function keys is explained below.

The first column of information below shows the BASE24 keys. The second column describes the functions that can be accomplished with these function keys.

Key	Description
F1	HI — Displays the high-level screen. When this function key is pressed from the low-level screen, the high-level screen is displayed. When this function key is pressed from the high-level screen, the low-level screen is displayed. The format of the data displayed on the low-level screen is dependent upon the settings on the Display/Alter Configuration Mode screen.
Shift-F2	HELP — Accesses the ATM Standard Display Help screen to explain the characters used on the low-level screen.

Screen Control Codes

The following Diebold screen control codes are supported by BASE24-atm. Refer to the vendor documentation for more information on the screen control codes

Code	Description
<i>"text"</i>	Displays text in ASCII format. The text must be enclosed in double quotation marks, and embedded quotes must be doubled.
<i>hh</i>	Displays a character in hexadecimal format from the current character set. The valid value of <i>hh</i> is 20 through 5F hexadecimal.
FF	Clears the screen and places the cursor in the home position. This should be the first code in any screen definition, except overlay screens.
HT	Displays the name found on Track 1 of the card at the current cursor location.
SI, <i>rc</i>	Sets the cursor to a particular position on the screen. The value <i>r</i> specifies the row number of the cursor with valid values of positions @ to O in Diebold's graphic format, representing rows 1 through 16. The value <i>c</i> specifies the column number of the cursor with valid values of positions @ to ? in Diebold's graphic format, representing columns 1 through 32.
SO, <i>n</i>	Overlays the current screen with another screen. Typically this code would be preceded by (and usually followed by) a set cursor (SI) code, unless the referenced screen does its own cursor positioning. The value <i>n</i> specifies the number of the screen to be inserted.
VT, <i>hh</i>	Displays an alternate character. The value <i>hh</i> is the hexadecimal format of the character, with valid values of 20 through 41 (hexadecimal).

Warning Messages

The following warning messages are specific to the Diebold Low-Level Display screen and are generated when something that is not recommended has been done. For institution-specific reasons, however, certain programming practices may cause these warning messages to be displayed. You can override these messages by pressing the **F1** key.

Clear Screen (FF) at Column XX Should be First Command

Where: XX = the number of the column

Cause: The FF code is embedded in the message. Thus, all preceding data in the message is erased.

Remedy: The FF code should be in the first column of the message.

Track1 Name (HT) at Column XX May Cause Wraparound on Display

Where: XX = the number of the column

Cause: The entry program does not know where the cursor is located. (This can occur if an embedded overlay exists.)

Remedy: You should return to the High-Level Standard Display screen after the update is complete to verify that the display looks correct. If the display does not look correct, return to the Low-Level Standard Display screen and move the HT character so that wraparound does not occur.

Text at Column XX May/Will Cause Wraparound on Display

Where: XX = the number of the column

Cause: The entry program does not know where the cursor is located. (This can occur if an embedded overlay exists.)

Remedy: You should return to the High-Level Standard Display screen after the update is complete to verify that the display looks correct. If the display does not look correct, return to the Low-Level Standard Display screen and move the text at column XX so that wraparound does not occur.

Typically the Last Display CMD Should "Home" the Cursor

Cause: The last code does not explicitly set the cursor and, thus, is left at the last line of display text.

Remedy: Typically, the cursor is set to return to the home position (SI,@@), unless data entry is expected.

Typically an Overlay Will Not Use Set Cursor as Last CMD

Cause: The screen currently displayed is an overlay and is set to position the cursor.

Remedy: Screens that call overlays normally control cursor positioning. If the screen that calls this overlay is set up otherwise, the warning message can be overridden safely.

Diebold State Table Screen

The Diebold State Table screen can be accessed by entering function code 2 on the Configurator Process screen. The Diebold State Table screen allows you to reconfigure screens and conditions that are associated with various ATM transaction states and groups.

Transaction processing at the ATM is controlled by the configuration of state tables. These state tables comprise a number of sequential instructions that provide the ATM with the information to perform financial and administrative transactions. Most states include a screen number and a next state number as part of the table entry.

When a terminal is in service and a transaction is requested, the state tables are executed. As each new state is entered, a screen is displayed at the ATM. The screen specifies the action that is required before the transaction can continue. When the specified action is taken, the transaction flow continues to the next state.

The Diebold State Table screen is shown below, followed by field descriptions. The items displayed on the screen vary according to the value of the state selected.

DIEBOLD STATE TABLE
Type A Card Read State
State Number 000 Group 0000

<state-specific data>

F1=CHANGE STATE TYPE	F2=READ F9=NEXT	F3=ADD F10=PRINT	F4=DELETE	F5=UPDATE F16=EXIT
----------------------	--------------------	---------------------	-----------	-----------------------

State Number — A number that is associated with the desired state. Valid values are 000 through 250 and 256 through 511.

Field Length: 3 numeric characters
 Required Field: Yes
 Default Value: 000

Group — The number used to identify a particular load group (that is, a collection of states, screens, tables, and parameters) that is downloaded to the ATM. Each group provides certain functionality, such as, Group 0 is the basic group, Group 1 provides fast cash transaction support, and so on. The group number of each ATM is assigned in the TERMINAL GROUP field of the TDF. Refer to the topic “High-Level Standard Display Screen” for a complete list of groups.

Field Length: 4 numeric characters
 Required Field: Yes
 Default Value: 0000

State Type — The state type associated with the value in the State Number field. BASE24-atm supports 16 state types that Diebold has defined. When a state type defined by Diebold is entered in this field, the necessary fields for that state type are listed on the screen. You can define additional state types, but the Diebold entry program does not check the data entered in those states to ensure that they meet the prescribed parameters. In addition, these additional state types are not listed on the cross-reference listings of states and screens, and states and states.

Field Length: 3 alphanumeric characters, followed by a system-protected text description
 Required Field: Yes
 Default Value: A

When a state type defined by Diebold is entered in this field, the necessary fields for that state type are listed on the screen. The state values defined by Diebold are listed in the following table.

Designation	Description
State A	Card Read state
State B	Pin Entry state
State C	Unlock Depository Vandal Door state
State D	Clear Keys state
State E	Select Function state

Designation	Description
State F	Dollar Entry state
State G	Cent Check state
State H	Information Entry state
State I	Transaction Request state
State J	Close Transaction state
State K	Indirect Next state
State L	Card Write state
State M	Pin Entry (Track3 Write) state
State N	Camera Control state
State O	Vandal Shield state
State ?	Withdrawal Area Sensor Check state (Diebold 912 only)

Miscellaneous Data Screen

The Miscellaneous Data screen can be accessed by entering function code 3 on the Diebold Configurator Process screen. Miscellaneous configuration data is loaded in Write Command III messages from the BASE24-atm Diebold 910/911 and 912 Device Handler processes to the terminal. The screen is used to enter the following types of data for each group.

- Depository printer or camera control parameters.
- Number of bad card reads before sending unsolicited status.
- Logical unit number (LUNO).
- Up to 15 timers for financial transactions. The exact number of timers that should be entered is dependent on the ATM model.

A group number is entered on this screen to read, add, change, or delete the miscellaneous data for the group indicated.

The Miscellaneous Data screen is shown below, followed by field descriptions.

```

                                MISCELLANEOUS DATA      Group 0000

Depository Printer/Automatic Camera Feature Parameter .....
Number of Bad Card Reads before Unsolicited Status .....
LUNO (Logical Unit Number) .....
ATM Timers:  0= Customer Keyboard Response Time (in seconds) ..
              1= Timeout Screen Display Time .....
              2= Close State Screen Display Time .....
              3= Communications Response Time .....
              4= Depository Response Time .....
              5= Currency Delivery Time .....
              6= Offline Response (ie, Poll Interval) Time ....
              7= Delivery Door Open Time .....
              8= Securomatic Response Time .....
              9= Card Removal Time .....
Optional Timers, -> 10= Withdrawal Door Time .....
based on ATM's  -> 11= Vandal Shield Time .....
model and/or   -> 12= Insert Document Time .....
installed      -> 13= Remove Document Time .....
options        -> 14= Reserved for Future .....

=====
                                F2=READ      F3=ADD      F4=DELETE      F5=UPDATE
                                F9=NEXT      F10=PRINT     F16=EXIT
  
```

Depository Printer/Automatic Camera Feature Parameter — Valid values for the Automatic Camera Feature field are as follows:

0 or 1 = Automatic
2 or 3 = Controlled

Field Length: 1 numeric character
Required Field: No
Default Value: None

Number Of Bad Card Reads Before Unsolicited Status — The number of bad card reads required before the ATM transmits an unsolicited status message. The default value for this DIT option is 005.

Field Length: 3 numeric characters
Required Field: No
Default Value: None

(LUNO) Logical Unit Number — The logical unit number of the ATM terminal. This number identifies a specific terminal to BASE24-atm.

Field Length: 3 numeric characters
Required Field: No
Default Value: None

ATM Timers — Indicates the time-out values for each of the timers listed. Refer to the appropriate Diebold vendor documentation for timer descriptions.

Field Length: 3 numeric characters
Occurs: 15 times
Required Field: No
Default Value: None

PIN Data Screen

The PIN Data screen can be accessed by entering function code 4 on the Diebold Configurator Process screen. This screen is used when the Diebold 910, 911, and 912 ATMs are not set to do DES PIN encryption.

The PIN Data screen is shown below.

The screenshot shows a terminal window titled "PIN DATA Group 0000". Inside, the text "Pin Data:" is displayed. Below this is a large rectangular area for data entry. At the bottom of the screen is a menu of function keys: F2=READ, F3=ADD, F4=DELETE, F5=UPDATE, F9=NEXT, F10=PRINT, and F16=EXIT. The menu is separated from the main area by a dashed line.

```
PIN DATA  Group 0000

Pin Data:

=====
F2=READ   F3=ADD   F4=DELETE  F5=UPDATE
F9=NEXT   F10=PRINT F16=EXIT
```

Pin Data — The PIN data entered using this screen contains three characters of algorithm location/identification and seven fields of offset index.

Field Length: 10 numeric characters
Required Field: Yes
Default Value: None

Financial Institution Table Screen

The financial institution table (FIT) can be accessed by entering function code 5 on the Diebold Configurator Process screen. It is used to maintain FIT data. These tables are used only when a Diebold 910, 911, and 912 ATM is set to do DES PIN encryption. The screen can be used to add a record to the Configuration File for each financial institution allowed to use the ATM (up to a maximum of 509 FIT records).

The Financial Institution Table screen is shown below, followed by field descriptions.

```

                                FINANCIAL INSTITUTION TABLE

                                FIT Entry Number 000   Group 0000

PIDDX: Offset to FIID on Card .....
PFIID: FIID Prefix .....
PSTDx: Offset to Indirect Next State .....
PAGDX: Offset to Algorithm Number on Card .....
PMXPN: Maximum PIN Entry Length .....
PCKLN: Number of Digits used for PIN Check .....
PINPD: PIN field Pad Character .....
PANDX: Offset to PIN Base Data on Card .....
PANLN: PIN Base Data Field Length .....
PANPD: PIN Base Data Pad Character .....
PRCNT: Offset to PIN Retry Count on Card .....
POFDX: Offset to Offset Field on Card .....
PDCTB: Decimalization Field for DES Algorithm ...
PEKEY: Encrypted PIN Key for DES Algorithm .....
PINDX: Information for Index Entries in FIT .....

=====
                                F2=READ   F3=ADD   F4=DELETE  F5=UPDATE
                                F9=NEXT   F10=PRINT  F16=EXIT

```

FIT Entry Number — The number associated with the financial institution table (FIT) entry group to be impacted.

Field Length: 3 numeric characters
Required Field: Yes
Default Value: 000

Group — The number of the group associated with the FIT entry group indicated in the FIT Entry Number field.

Field Length: 4 numeric characters

Required Field: Yes

Default Value: 0000

Display/Alter Configuration Mode Screen

The Display/Alter Configuration Mode screen can be accessed by entering function code 6 on the Diebold Configurator Process screen. The screen allows you to change the size of the screen displayed on the Diebold standard display screen and the type of data shown in low-level mode on that screen.

The Display/Alter Configuration Mode screen is shown below, followed by field descriptions.

```
+-----+
| This screen allows changes to some of ENTDIEB's internal control flags. |
| NOTE: You can return to this screen at any time, from any other screen, |
| by pressing F13-MENU. When changes (if any) are completed, you |
| can return to where you left off on the other screen by pressing |
| F16-EXIT, thus avoiding a return to the Menu screen. |
+-----+

The following flag is used when F3-CREATE-FILE is pressed to determine what
type of configuration file will be created:
  Default Configuration Mode ... 0= TABS-9000 (16 x 32 Display Screen)
                                1= MDS-1000 (20 x 40 Display Screen)
                                2= FUJITSU (16 x 32 Display Screen)

The following flag is used on the Low-Level Display Screen to control how
ESC-G(raphic) alternate characters are formatted when reading or printing:
  Low-Level Display Format ..... 0= Display as 2-digit Hex
                                1= Display as Ascii-equivalent
                                2= Display both Ascii and alternate
                                   characters as 2-digit Hex

F1=ENTER                                     F16=EXIT
```

Default Configuration Mode — Specifies the type of Configuration File to be created when a Configuration File is created from the Diebold Configurator Process screen. Valid values are as follows:

- 0 = The Configuration File is for TABS 9000 terminals. The screen display size for these terminals is 16 by 32.
- 1 = The Configuration File is for Diebold 1000-series terminals. The screen display size for these terminals is 20 by 40.
- 2 = The Configuration File is for Fujitsu ACT 4000, 6000, 6100, or 7000 terminals. The screen display size for these terminals is 16 by 32.

Field Length: 1 numeric character
 Required Field: Yes
 Default Value: None

Low-Level Display Format — A value indicating the type of screen display available when accessing the low-level mode on the Diebold standard ATM display screen. Valid values are as follows:

- 0 = Display alternate characters as 2-digit hexadecimal characters.
- 1 = Display alternate characters as ASCII equivalent.
- 2 = Display ASCII and alternate characters as 2-digit hexadecimal characters.

Field Length: 1 numeric character

Required Field: Yes

Default Value: None

Enhanced Display Library Definition Screen

The Enhanced Display Library Definition screen can be accessed by entering function code 7 on the Diebold Configurator Process screen. This screen is used only for Diebold 912 terminals.

This screen is used to define the dimensions (in blocks) of the characters within a given library. A library definition is only required for libraries made up of multiblock characters.

Library 1 is defined for English. Library 2 is reserved for future use.

For further information on the enhanced display feature including field descriptions for this screen, refer to the ***Diebold TABS Manual, #386***.

The Enhanced Display Library Definition screen is shown below.

```

      ENHANCED DISPLAY
      LIBRARY DEFINITION

      Library 1

      Top Screen Margin ..... 000
      Left Screen Margin ..... 000

      Character Block Height ..... 000
      Character Block Width ..... 000

      Top Character Margin ..... 000
      Left Character Margin ..... 000

      Character Height ..... 000
      Character Width ..... 000

      Character Rows per Screen .... 000
      Characters per Row ..... 000

=====
      F2=READ      F3=ADD      F4=DELETE  F5=UPDATE
      F9=NEXT      F10=PRINT   F16=EXIT
  
```

For further information on the enhanced display feature available with the Diebold 912 terminals, refer to Diebold's ***MDS Enhanced Character Supplement to the TABS 912/922 Programming Manual***.

Enhanced Display Character Definition Screen

The Enhanced Display Character Definition screen is used to define the characters for those libraries where each character is actually a composite of different subsets of preexisting characters (or blocks) pre-programmed into the programmable read only memory (PROM) solid state chip device (for example, each Chinese character as it appears on the screen is actually made up of four blocks).

This feature provides the ability to display non-English characters or symbols. This feature is used only for Diebold 912 terminals.

An enhanced character PROM with predefined character libraries is required to use this feature.

Warning: This Enhanced Display Entry screen must be used only to modify the Configuration File for Diebold 912 ATMs equipped with the MDS enhanced character generator PROM. For information on enhanced display entry screens used to modify the Configuration File for ATMs equipped with the Philips Enhanced Character Generator PROM, refer to appendix A.

The Enhanced Display Character Definition screen is shown below.

```

Character 00  Group 0000      ....+....1....+....2....+....3....+....4
Library Identifier 1          1  >
                               2  >
*****                      3  >
*   ENHANCED DISPLAY   *    4  >
*  CHARACTER DEFINITION  *    5  >
*****                      6  >
                               7  >
+-----+                   8  >
| Using blanks and        |  9  >
| asterisks, "draw" the   | 10  >
| character as it should  | 11  >
| appear on the CRT.      | 12  >
|                          | 13  >
| Make sure the height    | 14  >
| and width you use to    | 15  >
| draw the character      | 16  >
| match those defined in  | 17  >
| the Enhanced Display    | 18  >
| Library Definition.     | 19  >
+-----+                   20  >
                               ....+....1....+....2....+....3....+....4

F1=L0-LEVEL  F2=READ  F3=ADD  F4=DELETE  F5=UPDATE  F9=NEXT  F10=PRINT  F16=EXIT

```

This screen allows the design and modification of each character in a character set library. There are two Diebold entry program screens available for this purpose. The first screen displays a character, enlarged so that each pixel is one character on the screen. By using the facilities associated with this screen, you can modify the character as you wish it to appear.

The second screen, available by pressing the **F1** key from the first screen, allows the character definition to be edited in its native format, which is the hexadecimal representation of the character, treating each pixel as a bit that is either On or Off. Each character is then referenced by its index into the library.

ACI's default library indexes are, in part: A (Index 21), B (Index 22), ... Y (Index 39), and Z (Index 3). The reason for starting A at Index 21 is to support Track 1 read. Up to 240 characters can be defined per library.

Enhanced Display Screen Definition Screen

The Enhanced Display Screen Definition screen is used as a composing screen using characters defined in the Enhanced Display Character Definition Screen. This screen is only used for Diebold 912 and 922 terminals.

An enhanced character PROM with predefined character libraries is required to use this screen.

Warning: This Enhanced Display Screen Definition screen must be used only to modify the Configuration File for Diebold 912 ATMs equipped with the MDS enhanced character generator PROM. For information on enhanced display entry screens used to modify the Configuration File for ATMs equipped with the Philips Enhanced Character Generator PROM, refer to appendix A.

The Enhanced Display Screen Definition screen is shown below.

ENHANCED DISPLAY SCREEN DEFINITION Screen Number 000 Group 0000 Enter screen commands below, separating each command and each byte of data with a comma. Command syntaces are at right.	<pre> +-----+ FF=Clear Screen HT=Track1 Name SI=Set Cursor , <row> , <col> SO=Overlay Screen , <screen #> VT=Use Library , <library #> xx=Character Index +-----+ </pre>								
.....1.....2.....3.....4.....5.....6.....7.....+ 1 2 3 4 5 6 7 8 9 10 1.....2.....3.....4.....5.....6.....7.....+									
<table border="0" style="width: 100%;"> <tr> <td>F2=READ</td> <td>F3=ADD</td> <td>F4=DELETE</td> <td>F5=UPDATE</td> </tr> <tr> <td>F9=NEXT</td> <td>F10=PRINT</td> <td>F16=EXIT</td> <td></td> </tr> </table>		F2=READ	F3=ADD	F4=DELETE	F5=UPDATE	F9=NEXT	F10=PRINT	F16=EXIT	
F2=READ	F3=ADD	F4=DELETE	F5=UPDATE						
F9=NEXT	F10=PRINT	F16=EXIT							

The Enhanced Display Screen Definition screen is used to define each screen to be displayed by the ATM. Due to the limitation of the terminal on which the Diebold entry program is running, this function must be performed at the native command level, rather than the Diebold screen emulation level used by entering 1 and pressing the **F1** key from the Diebold Configurator Process screen. Therefore, caution should be used when modifying enhanced displays, since the actual

appearance of the screen when it is displayed by the Diebold ATM is not shown by the Diebold entry program. The same screen number restrictions apply to enhanced display screens as for standard 9000 display screens.

For further information on the enhanced display feature available with the Diebold 912 terminals, refer to the Diebold ***MDS Enhanced Character Supplement to the TABS 912/922 Programming Manual***.

Section 5

ATM Printer Support

Standard default receipt formats based on Diebold guidelines are provided with BASE24-atm. You can customize these default receipt formats. This section provides examples of default financial and administrative receipt formats as well as instructions on how to configure customized receipts. Refer to the vendor documentation for information on specifications for the physical properties and standard print formats supported by Diebold.

BASE24-atm Receipts

BASE24-atm can produce a printed receipt for the cardholder, which provides a tangible result of the electronic transaction that just took place. Receipts are also printed on the journal printer, so that the ATM owner has a record of the transaction.

Types of Printed Receipts

BASE24-atm can print receipts for the following transaction types:

- Financial transactions
- Administrative transactions

Financial Transactions

Each financial transaction has its own type of receipt, so that the information provided to the cardholder is tailored to the requirements of the transaction. A denied transaction also has its own type of receipt. Whether a receipt is printed can be determined by you at the terminal level, or you can allow the cardholders to decide if they want receipts.

Administrative Transactions

Administrative transactions are performed by institution personnel, to perform such tasks as replenishing currency. You can configure whether administrative receipts are printed in the TDF.

Configurable Receipts

Configurable receipts provide you with the ability to establish ATM receipt formats that are unique to the institution. In addition, this customization option supports the display of an informational or marketing message on the cardholder receipt, the use of flexible monetary symbols, and multiple language support. The configurable receipt option allows for configuring approved and denied financial transaction receipts as well as administrative receipts.

Receipt configuration is supported by device type, receipt group, transaction type, language code, card issuer FIID, *from* and *to* account type, and printer type. Receipt groups allow the use of a single receipt format for a group of ATMs rather than configuring receipts for each ATM. Receipt groups are specified for each terminal via Terminal Data File (TDF) screen 8. For further information on establishing receipt groups in the TDF, refer to the ***BASE24-atm Files Maintenance Manual***. Receipts can be configured for each transaction in each language that the device supports. Customized receipt formats are configured by maintaining records in the Receipt File (RCPT).

The total number of characters of printer data for all the blocks combined in one Function Command Message (that is, transaction response from BASE24-atm) cannot exceed 250. If this number is exceeded, the Device Handler process generates a log message and no receipt is printed.

There are 28 defined print lines on a receipt with 30 character positions per line. However, not all lines and character positions are available for machine printing. The receipt has 20 machine print lines and BASE24-atm uses 26 characters in width, for each line.

Receipt File

The Receipt File (RCPT) is the primary file used to configure receipts. It contains receipt templates that define the actual text and format to use for the various types of receipts.

During initialization, the Device Handler process reads the information in the RCPT into extended memory and then uses that information for creating receipts.

Note: Because the RCPT information is read into extended memory, the Device Handler process must be warmbooted every time the RCPT is modified for the modifications to take effect.

Initially, the RCPT contains default templates for each of the receipt types. You can use these default templates as they are, or you can modify the templates to print the information you want on the receipt.

Receipt Templates

Receipt templates are made up of a series of field markers and field type codes that define the information to be included on the receipt and the format of the receipt. Refer to the topics “Field Markers” and “Field Type Codes” for a description of field markers and field type codes that can be used in the receipt templates.

Individual receipt templates can be created for the various types of approved and denied financial transactions, and approved and denied administrative transactions. These same receipt templates can also be created in different languages.

To determine the text and format to use for a receipt, the Device Handler process scans a template to identify the field markers and field type codes and their positions in the template. Then, based on the field type codes, the Device Handler process retrieves the required information and formats the receipt information based on the field markers used and their positions in the template. Examples of the default templates provided in the RCPT for various transactions and examples of actual receipts created from these default templates are included later in this section.

Receipt templates are divided into three general categories, which are as follows:

- Receipt header templates control the information printed at the top of the receipt for all types of transactions.
- Receipt body templates control the information printed on the remainder of the receipt; body templates are specific to the type of transaction (for example, withdrawal or deposit, approved or denied, financial or administrative).
- Receipt trailer templates allow additional marketing information to be placed at the end of each financial transaction. Trailers are not used when the cardholder chains transactions and you allow multiple transactions to print on one receipt. Trailers can be multiple lines, however, the length of potential receipts with header, body, and trailer should be taken into consideration so that the number of lines does not exceed the maximum. Trailers are only allowed on financial transactions.

To correctly format receipt information for a transaction, a Device Handler process must be able to retrieve the appropriate templates from its RCPT information.

Receipt Template Keys

Each receipt template is one record in the RCPT, and the key fields for RCPT records are as follows: device type, receipt type, receipt group, language code indicator, FIID, *from* account type, *to* account type, and printer type. This group of key fields, described below, provides you with a great deal of flexibility in defining templates. For a comprehensive list of values that can be used in each of these key fields, refer to the ***BASE24-atm Files Maintenance Manual***.

Key Field	Description
DEVICE TYPE	Groups the RCPT records by type of device, allowing the Device Handler process to locate its records at initialization.
RECEIPT TYPE	Identifies the type of receipt template. Valid type codes have been defined by BASE24-atm.
RECEIPT GROUP	Allows the use of a single receipt template for a group of ATMs. Receipt groups are specified for each ATM on TDF screen 8. When selecting template data, the Device Handler process determines the group number assigned to an ATM in the TDF, and then searches for templates with the corresponding group. If an exact match cannot be found for the ATM group, the Device Handler process uses template records with asterisks in their group fields.
LANGUAGE CODE INDICATOR	Allows you to define the same receipt templates for more than one language. That is, a separate receipt template can be created in each of two languages supported by the ATM. A language indicator is sent to BASE24-atm from the ATM after the cardholder selects a preferred language for the transaction. The Device Handler process then uses this language indicator to select the templates to use. For example, you could configure one receipt template in English and one in Spanish in the RCPT. Each would be assigned a corresponding language indicator, which would allow the Device Handler process to select the appropriate template based on the language indicator sent from the ATM.

Key Field	Description
FIID	Allows a receipt to be formatted differently depending on who issued the card or owns the terminal. For instance, you may want different information printed for an on-us transaction than a not-on-us transaction.
FROM ACCOUNT TYPE	Allows a receipt to be formatted differently depending on the type of <i>from</i> account. For example, you can truncate savings account numbers differently than credit card numbers.
TO ACCOUNT TYPE	Allows a receipt to be formatted differently depending on the type of <i>to</i> account. For example, you can truncate savings account numbers differently than credit card numbers.
PRINTER TYPE	Can contain only the default value (*) for the Diebold 910/911 and 912 Device Handler processes.

Default Receipt Templates

Default receipt templates are provided in the RCPT for each type of transaction. These default receipt templates can be edited by you to customize specific types of information printed on the ATM receipts. Examples of the various default templates are described later in this section.

Multiple Languages

Receipt templates can be configured in more than one language. That is, a separate receipt template can be created in each of the two languages supported by each Diebold terminal for a withdrawal transaction. The language indicator is sent to BASE24-atm from the ATM after the cardholder selects a preferred language for the transaction. For example, by configuring one receipt template in English and one in Spanish and matching each receipt template to the appropriate language indicator in the RCPT, cardholder receipts can be printed in English or in Spanish.

The Device Handler process uses the eighth byte in the operations code buffer to determine the language to use in screen display and receipt printing.

Components of the Receipt Template

A number of components make up a receipt template. They are as follows:

- Field markers
- Field type codes
- Static text
- Directives

Field Markers

Field markers are special characters that identify the type of data to be used on a receipt or specify an action such as a line feed. The field markers that can be used with financial and administrative configurable receipt templates in the RCPT are described on the following pages.

The following table describes the standard field markers that can be used in receipt templates to define receipt text and format.

Code	Description
#	Indicates that a variable data field is to be inserted. The number symbol (#) is immediately followed by a field type code which indicates the type of data to be inserted. The possible field type codes are discussed below in the topic “Field Type Codes.”
>	Indicates that a line feed character (hexadecimal 0A) is to be inserted.
@	Indicates that a literal data string will follow.
?	Indicates that a directive statement will follow. A directive gives a command other than printing. The possible directives are discussed below in the topic “Directives.”

Field Type Codes

Field type codes correspond to predetermined pieces of fixed-length data which the Device Handler process retrieves from the BASE24-atm Standard Internal Message (STM), the TDF, or the C910NAMS (for 910 and 911 terminals) or C912NAMS (for 912 terminals). The field type codes that can be used in

configurable financial transaction receipt templates in the RCPT are described in the topic “Financial Transaction Field Type Codes”, and are divided into codes used by financial transactions and codes used by administrative transactions. The field type codes are divided as follows:

- Standard field type codes. These codes are used with all receipt types.
- Cartridge field type codes. These codes are used with administrative receipts only.
- Terminal totals field type codes. These codes are used with administrative receipts only.

Data names are provided to indicate the specific data fields from which the information is taken. For further information on the STM data fields, refer to the ***BASE24-atm Transaction Processing Manual***. For further information on the TDF data names, refer to the ***BASE24-atm Files Maintenance Manual*** (using the data name index) or refer to the DDLs.

Static Text

You can enter any text in the receipt template that you want to print on the receipt. Static text must be preceded by the at (@) symbol in the template. You must ensure that the length of the static text plus the length of any field type codes used does not exceed the maximum number of characters that can be printed on the receipt.

Special characters that are reserved for use with field markers (#, >, @, ?, and ^) can be printed in the text of the receipt by preceding them with the number (#) symbol. For instance, to print the greater than character (>), you would enter #> in the template.

Directives

A directive indicates a command other than printing. The IF and ENDIF directives allow one template to print different text and/or data fields depending upon a condition that is determined outside of the RCPT. The IF directive marks the beginning of a block of receipt template that will be printed if the condition is true; on the same line as the IF directive is the variable name of the condition to be tested. The ENDIF directive marks the end of the block. If the condition is not true, the block will be skipped, and the receipt will continue printing with the receipt line following the ENDIF directive.

Currently, FEES and NOFEES are the only valid condition values, and can only be used in financial receipt templates. When the FEES condition is on the same line as the IF directive, the following block will be printed if there is a surcharge fee assessed on the transaction. When the NOFEES condition is on the same line as the IF directive, the following block will be printed if there is no fee for the transaction (that is, surcharging is not configured, or surcharging is configured, but not applicable for this particular transaction).

An example of directives is described later in this section in the topic “Financial Transaction Receipts.” Refer to the Withdrawal from Checking, No Surcharge and Withdrawal from Checking, with Surcharge examples.

Financial Transaction Receipts

BASE24-atm supports both approved and denied transaction receipts. The following pages describe the default receipt templates that are used to configure approved financial transaction receipts as well as denied transaction receipts.

Default receipt templates are provided by ACI in English only regardless of the language indicator associated with the template. You must then modify the appropriate template if you want another language. For example, default receipt templates for language indicators 0 and 1 are configured in English. You must modify the templates to correspond to the languages supported, or add templates for additional languages you want to support. This section shows one example for each default receipt template in the RCPT.

The Device Handler process logs messages if receipt formatting errors occur when BASE24-atm transaction processing takes place. The Device Handler process also logs messages if the maximum number of lines per receipt or the print buffer is exceeded.

The ATM waits until each transaction is completed before printing the receipt. Multiple transactions can be printed on a receipt depending on the transaction receipt option selected in the TDF. The MAX LINES PER RCPT field on TDF screen 8 determines the number of transactions per receipt.

Balance inquiry information requested by the cardholder can be printed, displayed on the ATM screen, or both, depending on how the TDF and IDF are configured.

Optional Receipts

You can determine whether receipts are printed for each transaction by several methods. The first method allows you to control receipt and audit printing by transaction type, and by whether the transaction was approved or denied. You enter this information on TDF screen 8, which is a device-specific screen. See the ***BASE24-atm Files Maintenance Manual*** for information on this file.

The second method, only available with Diebold 910 and 911 terminals, allows the cardholder to make the final decision as to whether a receipt should be printed. If the cardholder chooses not to have a receipt printed, the audit receipt is still printed if your settings on TDF screen 8 indicate that it should print. Note that the cardholder's choice with the second method overrides your settings on TDF screen 8 with the first method.

Configuring Optional Receipts

The optional receipts function is determined by settings on TDF screen 7, which is also a device-specific screen. The OPTIONAL RECEIPTS field on TDF screen 7 has four values, as follows:

- 0 = Optional receipts are not supported.
- 1 = The cardholder selects optional receipts if the printer fails. If the printer is working, your settings on TDF screen 8 are used.
- 2 = The cardholder selects optional receipts on all approved transactions. If the transaction is denied, your settings on TDF screen 8 are used.
- 3 = The cardholder selects optional receipts on all approved and denied transactions.

Two additional fields on TDF screen 7, PRINTER OPERABLE GROUP and PRINTER INOPERABLE GROUP, determine how often and where in the flow the cardholder is asked about optional receipts. The inoperable group contains the series of screens that are downloaded when the terminal sends a status indicating that the printer is not working. The operable group contains the series of screens that are downloaded initially with group 0 and group 300, and then downloaded again when the printer is once again working. One of the following options is used, depending on the values that you enter in the two group fields:

- The cardholder selects once at beginning of the session whether the receipt will be printed (decision applies to all chained transactions).
- The cardholder selects during each transaction whether the receipt will be printed (decision is for that transaction only).

The groups that ACI provides in a sample Configuration File for the optional receipts function are as follows (the options are set in OPTIONAL RECEIPTS field as described above):

- Group 300 = The basic group for the optional receipts function.
- Group 301 = The printer currently is operational. The cardholder selects optional receipts once for all chained transactions.
- Group 302 = The printer currently is inoperable. The cardholder selects optional receipts once for all chained transactions (option 1, 2, or 3).
- Group 303 = The printer currently is operational. The cardholder selects optional receipts once for all chained transactions, only if the printer becomes inoperable (option 1).

- Group 311 = The printer currently is operational. The cardholder selects optional receipts before each transaction.
- Group 312 = The printer currently is inoperable. The cardholder selects optional receipts before each transaction (option 1, 2, or 3).
- Group 313 = The printer currently is operational. The cardholder selects optional receipts before each transaction, only if the printer becomes inoperable (option 1).

You can use the states, financial institution tables (FIT), and screens to create subsets of cardholders that use various settings of the optional receipts functionality. However, the sample Configuration File provided by ACI only shows settings for all cardholders. Refer to section 4 for a description of groups, states, FITs, and screens.

Transaction Log File

The Transaction Log File (TLF) contains a flag indicating whether a receipt was requested. Valid values for the receipt option flag are as follows:

- 0 = The cardholder was not asked whether a receipt was required.
- 1 = A receipt was requested.
- 2 = A receipt was not printed because the printer has faulted.
- 3 = A receipt was not printed because the cardholder requested that one not be printed.

You can view this flag on the TLF. Refer to the ***BASE24-atm Files Maintenance Manual*** for information on how to peruse the TLF. However, this flag is not on any of the printed BASE24 reports.

Samples of Optional Receipt Settings

The following topics show how the different settings of the TDF work together to provide six different scenarios for optional receipts. These settings determine when the optional receipts function is used, and how often the cardholders are asked about whether they want to complete a transaction without a printed receipt.

Scenario 1—Ask Once per Session Only if Printer Fails

To ask the cardholder only once per session whether a receipt is printed if the printer fails, enter the following values in the TDF fields:

OPTIONAL RECEIPT = 1
PRINTER OPERABLE GROUP = 0303
PRINTER INOPERABLE GROUP = 0302

If the printer is working, the settings on TDF screen 8 are used. If the printer faults in the middle of a series of chained transactions, the cardholder is asked “Continue without a receipt?” during each transaction. If the cardholder selects yes, the transaction is allowed without a receipt. If the cardholder selects no, the Main Menu is displayed, and the cardholder can select another transaction. Once the cardholder has ended the session, the printer inoperable group is downloaded.

A subsequent cardholder is asked at the beginning of the session, “Continue without a receipt?” If the cardholder selects yes, then all chained transactions are allowed without a receipt. If the cardholder selects no, the card is returned and the session is ended.

Once the printer is serviced and the terminal exits the maintenance mode, the printer operable group is downloaded.

Scenario 2—Ask Each Transaction Only if Printer Fails

To ask the cardholder during each transaction whether a receipt is printed if the printer fails, enter the following values in the TDF fields:

OPTIONAL RECEIPT = 1
PRINTER OPERABLE GROUP = 0313
PRINTER INOPERABLE GROUP = 0312

If the printer is working, the settings on TDF screen 8 are used. If the printer faults in the middle of a series of chained transactions, the cardholder is asked “Continue without a receipt?” during each transaction. If the cardholder selects yes, the transaction is allowed without a receipt. If the cardholder selects no, the Main Menu is displayed, and the cardholder can select another transaction. Once the cardholder has ended the session, the printer inoperable group is downloaded.

A subsequent cardholder is asked during each transaction, “Continue without a receipt?” If the cardholder selects yes, the current transaction is allowed without a receipt. If the cardholder selects no, the Main Menu is displayed, and the cardholder can select another transaction.

Once the printer is serviced and the terminal exits the maintenance mode, the printer operable group is downloaded.

Scenario 3—Ask Once per Session on Approved Transactions

To ask the cardholder once per session whether a receipt is printed on all approved transactions, enter the following values in the TDF fields:

OPTIONAL RECEIPT = 2
PRINTER OPERABLE GROUP = 0301
PRINTER INOPERABLE GROUP = 0302

When the printer is working, the cardholder is asked once at the beginning of the session, “Do you want a receipt?” If the cardholder selects yes, the receipts for all approved chained transactions are printed. There is one exception to the “yes” selection; if the transaction is denied, the settings on TDF screen 8 are used.

If the cardholder selects no, receipts are not printed for all approved chained transactions. There are two exceptions to the “no” selection:

- If the transaction can only complete a partial dispense, a receipt will always be printed.
- If the transaction is denied, the settings on TDF screen 8 are used.

If the printer faults in the middle of a series of chained transactions, the cardholder is asked “Continue without a receipt?” during each transaction. If the cardholder selects yes, the transaction is allowed without a receipt. If the cardholder selects no, the Main Menu is displayed, and the cardholder can select another transaction. Once the cardholder has ended the session, the printer inoperable group is downloaded.

A subsequent cardholder is asked at the beginning of the session, “Continue without a receipt?” If the cardholder selects yes, then all chained transactions are allowed without a receipt. If the cardholder selects no, the card is returned and the session is ended.

Once the printer is serviced and the terminal exits the maintenance mode, the printer operable group is downloaded.

Scenario 4—Ask Once per Session on Approved and Denied Transactions

To ask the cardholder once per session whether a receipt is printed on all approved and denied transactions, enter the following values in the TDF fields:

OPTIONAL RECEIPT	= 3
PRINTER OPERABLE GROUP	= 0301
PRINTER INOPERABLE GROUP	= 0302

When the printer is working, the cardholder is asked once at the beginning of the session, “Do you want a receipt?” If the cardholder selects yes, the receipts for all approved and denied chained transactions are printed. If the cardholder selects no, receipts are not printed for all approved or denied chained transactions. There is one exception to the no selection; if the transaction can only complete a partial dispense, a receipt will always be printed.

If the printer faults in the middle of a series of chained transactions, the cardholder is asked “Continue without a receipt?” during each transaction. If the cardholder selects yes, the transaction is allowed without a receipt. If the cardholder selects no, the Main Menu is displayed, and the cardholder can select another transaction. Once the cardholder has ended the session, the printer inoperable group is downloaded.

A subsequent cardholder is asked at the beginning of the session, “Continue without a receipt?” If the cardholder selects yes, then all approved and denied chained transactions are allowed without a receipt. If the cardholder selects no, the card is returned and the session is ended.

Once the printer is serviced and the terminal exits the maintenance mode, the printer operable group is downloaded.

Scenario 5—Ask Each Transaction on Approved Transactions

To ask the cardholder during each transaction whether a receipt is printed on all approved transactions, enter the following values in the TDF fields:

OPTIONAL RECEIPT = 2
PRINTER OPERABLE GROUP = 0311
PRINTER INOPERABLE GROUP = 0312

When the printer is working, the cardholder is asked during each transaction, “Do you want a receipt?” If the cardholder selects yes, the receipt for the approved transaction is printed. There is one exception to the “yes” selection; if the transaction is denied, the settings on TDF screen 8 are used.

If the cardholder selects no, a receipt is not printed for the approved transaction. There are two exceptions to the “no” selection:

- If the transaction can only complete a partial dispense, a receipt will always be printed.
- If the transaction is denied, the settings on TDF screen 8 are used.

If the printer faults in the middle of a series of chained transactions, the cardholder is asked “Continue without a receipt?” during each transaction. If the cardholder selects yes, the transaction is allowed without a receipt. If the cardholder selects no, the Main Menu is displayed, and the cardholder can select another transaction. Once the cardholder has ended the session, the printer inoperable group is downloaded.

A subsequent cardholder is asked during each transaction, “Continue without a receipt?” If the cardholder selects yes, the approved transaction is allowed without a receipt. If the cardholder selects no, the Main Menu is displayed, and the cardholder can select another transaction.

Once the printer is serviced and the terminal exits the maintenance mode, the printer operable group is downloaded.

Scenario 6—Ask Each Transaction on Approved and Denied Transactions

To ask the cardholder during each transaction whether a receipt is printed on all approved and denied transactions, enter the following values in the TDF fields:

OPTIONAL RECEIPT = 3
PRINTER OPERABLE GROUP = 0311
PRINTER INOPERABLE GROUP = 0312

When the printer is working, the cardholder is asked during each transaction, “Do you want a receipt?” If the cardholder selects yes, the receipt for the approved or denied transaction is printed. If the cardholder selects no, a receipt is not printed for the approved or denied transaction. There is one exception to the “no” selection; if the transaction can only complete a partial dispense, a receipt will always be printed.

If the printer faults in the middle of a series of chained transactions, the cardholder is asked “Continue without a receipt?” during each transaction. If the cardholder selects yes, the transaction is allowed without a receipt. If the cardholder selects no, the Main Menu is displayed, and the cardholder can select another transaction. Once the cardholder has ended the session, the printer inoperable group is downloaded.

A subsequent cardholder is asked during each transaction, “Continue without a receipt?” If the cardholder selects yes, the transaction is allowed without a receipt. If the cardholder selects no, the Main Menu is displayed, and the cardholder can select another transaction.

Once the printer is serviced and the terminal exits the maintenance mode, the printer operable group is downloaded.

Sample Financial Transaction Receipt

The Device Handler process sends to the terminal all of the information needed to print a financial transaction receipt.

The table below describes the standard lines used on Diebold ATM default financial receipts. Because receipts are configurable, however, the information on each receipt line could be different.

Line	Description
1	The date of the transaction, the time the transaction occurred, and the first 8 digits of the 16-digit terminal ID of the Diebold ATM.
2	Blank
3	The card number that was used to initiate the transaction.
4	Blank
5	ATM location; obtained from the TDF.
6–7	Blank
8	A label of RECORD NO. and the terminal sequence number assigned to this transaction.
9	The transaction type that has been performed and the amount of the transaction.
10	Blank
11	The account number of the affected account. If funds are being withdrawn <i>from</i> the affected account the account number is preceded by the label FR. If funds are being deposited <i>to</i> the affected account the account number is preceded by the label TO. If a two-sided transaction such as transfer, inquiry, or payment has been performed, the affected <i>from</i> account number is printed on this line, preceded by the label FR.
12	A space reserved for the LEDGER BALANCE field, if needed. The Diebold ATM prints the ledger balance of the affected account immediately following the LEDGER BALANCE field for deposit, withdrawal, or inquiry transactions. If a two-sided transaction such as transfer or payment has been performed, the affected <i>to</i> account number is printed on this line, preceded by the label TO.

Line	Description
13	A space reserved for the AVAILABLE BALANCE field, if needed. The Diebold ATM prints the available balance of the affected account immediately following the AVAILABLE BALANCE field for withdrawal, deposit, or inquiry transactions.

The following is a sample receipt that demonstrates the standard lines for an Diebold receipt.

```

1      12/15/96  08:30   S1AC9100
2
3      376666000000000001-000
4
5      330 S. 108TH AVE
6
7
8      RECORD NO.           83
9      DEPOSIT              $20.00
10
11     TO 5555566666777778888
12     LEDGER BAL           $123.45
13     AVAIL BAL            $678.90
14
15

```

```

          1      2
1234567890123456789012345678

```

Financial Transaction Field Type Codes

The following table describes the standard field type codes that are used in receipt templates. Each field type code corresponds to a predetermined piece of fixed-length data that the Device Handler process retrieves from the BASE24-atm Standard Internal Message (STM), the TDF, or the C910NAMS (for 910 and 911 terminals) or C912NAMS (for 912 terminals). Data names are included in parentheses after each field type code description to indicate the specific data field from which the information is taken. This list includes field type codes that are not used in the default receipt templates, but which you can use to customize the receipt templates.

For further information on the STM data fields, refer to the **BASE24-atm Transaction Processing Manual**. For further information on the TDF data names, refer to the **BASE24-atm Files Maintenance Manual** (using the data name index) or refer to the DDLs.

Code	Description
A1	The transaction amount (STM.RQST.AMT-1), for instance, the withdrawal amount or deposit amount. In the case of a withdrawal with a surcharge fee, A1 contains the withdrawal amount plus the surcharge (which is the amount that will be deducted from the cardholder's account).
A2	The transaction amount (STM.RQST.AMT-2). This amount is the ledger balance for most transactions.
A3	The transaction amount (STM.RQST.AMT-3). This amount is the available balance for most transactions.
AB	The available balance (STM.RQST.AMT-3). This code does not print the text, "AVAIL BAL" (available balance), on the receipt. Use code XA below to print the text on the receipt. If this code is used on a balance inquiry receipt, the available balance is only printed if a response code 0 is returned from the BASE24-atm Authorization process and the CUSTOMER BALANCE INFO fields in the Institution Definition File (IDF) and the TDF indicate that Amount 3 (STM.RQST.AMT-3) is to be displayed.
CN	The card number (ATM.D910.CRD-NUM or ATM.D912.CRD-NUM) and member number (STM.RQST.MBR-NUM) from the ATM card. To print card number only, see the CX field type code below.

Code	Description
CXL, <i>l,r,c</i>	The card number from the ATM card (ATM.D910.CRD-NUM or ATM.D912.CRD-NUM). The member number is not printed. The three variables (<i>l</i>, <i>r</i>, and <i>c</i>) are used to control the truncation of the card number. If the variables are not present, the card number will be printed in its entirety. Refer to the topic “Account Number Truncation” later in this section for additional information about these variables.
DA	The transaction date. (STM.TRAN-DAT—MM/DD/YY or DD/MM/YY) The date displayed is dependent on the value specified in the DATE-DISPLAY-TYPE param in the Logical Network Configuration File (LCONF).
FA	The <i>from</i> account type (for example, CK, SV, CC). The account type printed on the receipt depends on the language indicator and the value of STM.FROM-ACCT-TYP. The Device Handler process matches account type numbers to a two-character account type description in the ACCT^TYP^TBL^G table in the C910NAMS (for 910 and 911 terminals) or C912NAMS (for 912 terminals).
FD	The <i>from</i> account description (for example, CHECKING, SAVINGS, CREDIT CARD). The account description printed on the receipt depends on the language indicator and the value of STM.FROM-ACCT-TYP. The Device Handler process retrieves the account description in the APPL^TBL^G table in the C910NAMS (for 910 and 911 terminals) or C912NAMS (for 912 terminals).
FRL, <i>l,r,c</i>	The <i>from</i> account number (STM.RQST.FROM-ACCT). If the STM.RQST.FROM-ACCT is all blanks or zeros, the Device Handler process uses the account description from the APPL^TBL^G table in the C910NAMS (for 910 and 911 terminals) or C912NAMS (for 912 terminals). The value of STM.RQST.FROM-ACCT-TYP and the language indicator determine which table entry is used. The three variables (<i>l</i>, <i>r</i>, and <i>c</i>) are used to control the truncation of the account number. If the variables are not present, the account number will be printed in its entirety. Refer to the topic “Account Number Truncation” later in this section for additional information about these variables.

Code	Description
LB	The ledger balance (STM.RQST.AMT-2). This code does not print the text, “LEDGER BAL” (ledger balance), on the receipt. Use code XL below to print the text on the receipt. If this code is used on a balance inquiry receipt, the ledger balance is printed only if a response code of 0 is returned from the BASE24-atm Authorization process and the CUSTOMER BALANCE INFO fields in the IDF and the TDF indicate that Amount 2 (STM.RQST.AMT-2) is to be displayed.
MB	Cardholder member number. (STM.RQST.MBR.NUM)
OL	The number of the log only transaction selected (for example, Log Only #1.) Four Log Only transactions are supported. If <i>from</i> account is 01, then a 1 is printed on receipt; if 02, a 2; if 03, a 3; and if 04, a 4.
RC	The response code. (STM.RQST.RESP)
RT	The response text. This code is used with denied transaction receipts. The printed receipt text depends on the value of STM.RQST.RESP. This response code is mapped to an internal response code specific to this Device Handler process. The Device Handler process searches the RESPONSE^TEXT^G table in the C910NAMS (for 910 and 911 terminals) or C912NAMS (for 912 terminals) to find a match on the internal response code. The corresponding text is then moved to the print buffer. The RESPONSE^TEXT^G table is configurable; refer to the topic “Response Code Text Table for Financial Receipts” later in this section.
S1	In the case of a withdrawal with a surcharge fee, this code is the amount to be dispensed, without including the surcharge amount (STM.RQST.AMT-1 – TRAN-FEE). This code only can be used on 910 and 911 terminals.
S2	In the case of a withdrawal with a surcharge fee, this code is the amount of the surcharge (the TRAN-FEE field, from the Surcharge Data token). This code only can be used on 910 and 911 terminals.
SQ	The 4-digit BASE24 sequence number assigned to the transaction. (TDF.TRAN-SEQ-NUM)
T1	The time the transaction took place—24 Hour HH:MM. (STM.TRAN-TIM)

Code	Description
T2xx/yy	The time the transaction took place (STM.TRAN-TIM). Time is displayed in a 12-hour format as opposed to the 24-hour format. If the time is less than 1200, the time appears as HH:MMxx. If the time is greater than or equal to 1200, the time appears as HH:MMyy where HH has valid values of 1 through 12. xx and yy are the configurable variable length symbols for a.m. and p.m., respectively. The format of this field type code is T2xx/yy. You can optionally insert spaces before xx and yy if you do not want the text to directly follow the time. yy is terminated by either the next field marker code (#, >, @, or ?), or by the end of the template.
TA	The <i>to</i> account type (for example, CK, SV, CC). The account type printed on the receipt depends on the language indicator and the value of STM.TO-ACCT-TYP. The Device Handler process matches account type numbers to a two-character account type description in the ACCT^TYP^TBL^G table in the C910NAMS (for 910 and 911 terminals) or C912NAMS (for 912 terminals).
TC	The city where the terminal is located (STM.RQST.TERM-CITY).
TD	The <i>to</i> account description (for example, CHECKING, SAVINGS, CREDIT CARD). The account description printed on the receipt depends on the language indicator and the value of STM.TO-ACCT-TYP. The Device Handler process retrieves the account description in the APPL^TBL^G table in the C910NAMS (for 910 and 911 terminals) or C912NAMS (for 912 terminals).
TE	All 16 bytes of the terminal ID (STM.TERM-ID).
TL	The terminal location (STM.RQST.TERM-NAME-LOC).
TM	The first 8 bytes of the terminal ID (STM.TERM-ID).
TN	The name of the terminal owner (STM.RQST.TERM-OWNER-NAME).

Code	Description
TOL, <i>r,c</i>	The <i>to</i> account number (STM.RQST.TO-ACCT). If the STM.RQST.TO-ACCT is all blanks or zeros, the Device Handler process uses the account description from the APPL^TBL^G table in the C910NAMS (for 910 and 911 terminals) or C912NAMS (for 912 terminals). The value of STM.RQST.TO-ACCT-TYP and the language indicator determine which table entry is used. The three variables (<i>l</i>, <i>r</i>, and <i>c</i>) are used to control the truncation of the account number. If the variables are not present, the account number will be printed in its entirety. Refer to the topic “Account Number Truncation” later in this section for additional information about these variables.
TS	The state where the terminal is located. (STM.RQST.TERM-ST-X)
TT	The transaction type. The transaction type printed on the receipt depends on the value of STM.TRAN-CDE. The Device Handler process searches the TRAN^TBL^G table in the C910NAMS (for 910 and 911 terminals) or C912NAMS (for 912 terminals) to find a match on the transaction code, then the corresponding text is moved to the print buffer.
XA	The text “AVAILABLE BAL” (available balance) that is printed on the receipt. This text is configured in the LEDG^AVAIL^TBL^G table in the C910NAMS (for 910 and 911 terminals) or C912NAMS (for 912 terminals). This code does not call the amount of the ledger balance. Use code AB (above) to print the available balance amount.
XL	The text “LEDGER BAL” (ledger balance) that is printed on the receipt. This text is configured in the LEDG^AVAIL^TBL^G table in the C910NAMS (for 910 and 911 terminals) or C912NAMS (for 912 terminals). This code does not call the amount of the ledger balance. Use code LB (above) to print the ledger balance amount.

Account Number Truncation

Three of the field type codes listed above have additional variables that allow the account number or card number to be truncated. This feature can help prevent fraud by making it more difficult for an unscrupulous person to use a carelessly discarded receipt to gain access to an account.

The three field type codes are CX (card number without member number), FR (*from* account number), and TO (*to* account number). The three variables that can be appended to these field type codes are *l*, a two-character numeric code which indicates the number of the leftmost digits that are printed; *r*, a two-character numeric code which indicates the number of the rightmost digits that are printed; and *c*, a one-alphanumeric character code which indicates the pad character, or the character used to fill the remaining digits. If these three variables are not present, the card or account number will be printed in its entirety.

Note that you can use a blank space as a pad character. To use a reserved special character, (#, @, and >), precede it with the number symbol (#). For example, to print a > symbol, enter it in the template as #>.

Note that if you do truncate one or more of the numbers, this will not affect how the number is stored internally; the complete number will be shown on the TLF and reports.

The following examples use this 19-digit primary account number (PAN):
1123581321345589144

Field Type Code in Receipt Template	PAN as Printed on Receipt
#CX00, 04, %	%%%%%%%%%%%%%%9144
#CX06, 04, #@	112358@@@@@@@@@9144
#CX06, 00, X	112358XXXXXXXXXXXXX

Monetary Symbols

The amount field type codes (A1, A2, A3, AB, LB, S1, and S2) can have a monetary symbol (such as \$, ¥, or £) that prints to the receipt. You can enter the monetary symbol either manually or have it determined automatically by the

Device Handler process. To enter the symbol manually, type the appropriate symbol after each occurrence of the field type code. Any text after the code will be considered part of the monetary symbol until the next field marker is encountered.

The monetary symbol that is used can be determined automatically by the use of a table in the C910NAMS or C912NAMS. For the symbol to be determined automatically, you must enter CC\$ after the field type code. The symbol is determined by the currency code table (RCPT^MONEY^SYM^TBL^G) in the C910NAMS (for 910 and 911 terminals) or C912NAMS (for 912 terminals). This table contains an entry for each monetary system for which you want a special symbol to be printed. Each entry in the table contains the country code, the number of characters required in the sequence to send the special symbol(s) to the receipt printer (including escape codes), the number of characters taken up on the receipt as a result of the sequence, and the 10-character sequence that prints the proper monetary symbol. If a country code is encountered that is not in the table, the 3-character ISO currency code will be used (for example, USD for US dollars). If neither the country code nor the ISO currency code exist in the table, no symbol will be printed.

Transaction Receipt Header

The receipt header is printed at the top of the receipt for all financial transactions. The Device Handler process uses the header receipt template (the HDR receipt type in the RCPT) to format the receipt header for all transactions. As described earlier in this section, the process uses the device type, receipt group, transaction type, language code, card issuer FIID, from and to account type, and printer type from the RCPT to find the corresponding header receipt template.

Once this step is completed, the Device Handler process moves to the first character of the header receipt template and uses the field markers and field type codes described earlier in this section to scan the receipt template to determine the receipt text.

The following is a representation of the default entry in the header receipt template (receipt type HDR) in the RCPT for the Device Handler process.

Transaction Type	Receipt Type	Default RCPT Template Data
General Header	HDR	#DA #T1 #TM>>> #CN>> TL>>>

Transaction Receipt Trailer

A trailer or footer can also be added to a receipt to contain marketing information, or other information you want to print at the bottom of all financial receipts. You can choose not to use a trailer by not having a TRLR receipt type.

So that the trailer is not printed multiple times on one receipt, the trailer is ignored if the cardholder is chaining transactions and you have the TRANSACTIONS ON RECEIPT field on TDF screen 8 set to 0, for multiple transactions per receipt.

An example of the trailer receipt (receipt type TRLR) is not given, because it is specific to each institution and no default template is included in the RCPT.

Standard Receipt Samples

The following pages describe the default receipt templates provided in the RCPT. Default templates are provided in English only regardless of the language indicator associated with template. For example, default receipt templates for language indicators 0 and 1 are configured in English.

The following pages show default financial receipts for the following transactions:

- Withdrawal from checking (Diebold 912)
- Withdrawal from checking, no surcharge (Diebold 910 and 911)
- Withdrawal from checking, with surcharge (Diebold 910 and 911)
- Deposit to savings
- Transfer from checking to savings
- Checking account inquiry
- Credit payment from checking
- Payment enclosed
- Message enclosed
- Log only
- PIN change
- Denied financial transactions

Withdrawal from Checking, Diebold 912

The receipt type WD is used for withdrawal transactions. The Diebold default receipt template is shown below, followed by a graphic example of the default receipt for a withdrawal from checking account transaction. This template only applies to Diebold 912 terminals. Note that in the case of a partial dispense, the actual amount dispensed is printed, but no balance information is printed.

```
@RECORD NO.      #SQ>
@WITHDRAWAL#A1$>
@FR  #FR>>
#XL#LB$>
#XA#AB$>>
```

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19

12/15/96 15:39 S1AC9100

376666000000000001-000

330 S. 108TH AVE

RECORD NO. 838
WITHDRAWAL \$20.00
FR 1111122222333334444

LEDGER BAL \$123.45
AVAIL BAL \$678.90

1 2
123456789012345678901234567

Withdrawal from Checking, No Surcharge, Diebold 910/911

The receipt type WD is used for withdrawal transactions. The Diebold default receipt template is shown below, followed by a graphic example of the default receipt for a withdrawal from checking account transaction that does not have an added surcharge fee. This template only applies to Diebold 910 and 911 terminals. Note that in the case of a partial dispense, the actual amount dispensed is printed, but no balance information is printed.

```
@RECORD NO.      #SQ>
?IF NOFEES
@WITHDRAWAL#A1$>
?ENDIF
?IF FEES
@WITHDRAWAL#S1$>
@FEE             #S2$>
@TOTAL           #A1$>
?ENDIF
@FR  #FR>>
#XL#LB$>
#XA#AB$>>
```

```
1      12/15/96  08:30   S1AC9100
2
3      37666600000000001-000
4
5      330 S. 108TH AVE
6
7
8      RECORD NO.           570
9      WITHDRAWAL           $20.00
10     FR      013333
11
12     LEDGER BAL           $3412.00
13     AVAIL BAL           $1760.00
14
15
16
17
18
19
```

```
1      2
123456789012345678901234567
```

Withdrawal from Checking, with Surcharge, Diebold 910/911

The following is a graphic example of the Diebold 910 and 911 default receipt format for a withdrawal from checking account transaction that has an added surcharge fee. Note that this receipt is produced using the same template as the example on the previous page. Also note that in the case of a partial dispense, the actual amount dispensed is printed, but no balance information is printed.

1	12/15/96	08:30	S1AC9100
2			
3	376666000000000001-000		
4			
5	330 S. 108TH AVE		
6			
7			
8	RECORD NO.	571	
9	WITHDRAWAL	\$20.00	
10	FEE	\$1.00	
11	TOTAL	\$21.00	
12	FR	013333	
13			
14	LEDGER BAL	\$3412.00	
15	AVAIL BAL	\$1760.00	
16			
17			
18			
19			
20			

1 2
123456789012345678901234567

Deposit to Savings Account

The receipt type DP is used for deposit transactions. The Diebold default receipt template is shown below, followed by a graphic example of the default receipt for a deposit to savings account transaction. The account type may be printed on the receipt, instead of the cardholder account number, depending on the authorization method used and whether the account number is available to BASE24-atm.

```
@RECORD NO.      #SQ>
@DEPOSIT         #A1$>>
@TO  #TO>
#XL#LB$>
#XA#AB$>>
```

```
1  12/15/96  08:30  S1AC9100
2
3  376666000000000001-000
4
5  330 S. 108TH AVE
6
7
8  RECORD NO.      83
9  DEPOSIT         $20.00
10
11 TO 5555566666777778888
12 LEDGER BAL      $123.45
13 AVAIL BAL      $678.90
14
15
16
17
18
19
20
```

```
1 2
1234567890123456789012345678
```

Transfer from Checking to Savings

The receipt type TR is used for transfer transactions. The Diebold default receipt template is shown below, followed by a graphic example of the default receipt for a transfer from checking to savings transaction. Cardholder account numbers may be printed on the receipt, instead of the account type, depending on the authorization method used and whether the account number is available to BASE24-atm.

```
@RECORD NO.      #SQ>
@TRANSFER      #A1$>
@FR  #FR>
@TO  #TO>>
```

```
1  12/16/96 09:58   S1AC9100
2
3  376666000000000001-000
4
5  330 S. 108TH AVE
6
7
8  RECORD NO.      83
9  TRANSFER      $100.00
10 FR 5555566666777778888
11 TO 1111122222333334444
12
13
14
15
16
17
18
```

```
1 2
1234567890123456789012345678
```

Checking Account Inquiry

The receipt type INQ is used for inquiry transactions. The Diebold default receipt template is shown below, followed by a graphic example of the default receipt for a balance inquiry from checking transaction. The cardholder account number may be printed on the receipt, instead of the account type, depending on the authorization level used and whether the account number is available to BASE24-atm.

```
@RECORD NO.      #SQ>
@INQUIRY>
@FR  #FR>>
#XA#LB$>
#XA#AB$>>
```

1	12/16/96	12:42	S1AC9100
2			
3	376666000000000001-000		
4			
5	330 S. 108TH AVE		
6			
7			
8	RECORD NO.	48	
9	INQUIRY		
10	FR 5555566666777778888		
11			
12			
13			
14			
15	LEDGER BAL	\$123.45	
16	AVAIL BAL	\$678.90	
17			
18			
19			
20			

1	2
1234567890123456789012345678	

Credit Account Payment from Checking Account

The receipt type PY is used for payment transactions. The Diebold default receipt template is shown below, followed by a graphic example of the default receipt for a credit account payment made from a checking account. The cardholder account number may be printed on the receipt, instead of the account type, depending on the authorization method used and whether the account number is available to BASE24-atm.

```
@RECORD NO.      #SQ>
@PAYMENT    #A1$>
@FR    #FR>
@TO    #TO>>
```

```
1      12/17/96 12:44      S1AC9100
2
3      376666000000000001-000
4
5      330 S. 108TH AVE
6
7
8      RECORD NO.              50
9      PAYMENT                  $100.00
10     TO 5555566666777778888
11     FR 1111222223333344444
12
13
14
15
16
17
18
19
```

```
1      2
1234567890123456789012345678
```

Payment Enclosed

The receipt type PYEN is used for payment enclosed transactions. The Diebold default receipt template is shown below, followed by a graphic example of the payment enclosed transaction.

@RECORD NO. #SQ>
@PAYMT ENCL#A1\$>>

1	12/15/96	08:30	S1AC9100
2			
3	376666000000000001-000		
4			
5	330 S. 108TH AVE		
6			
7			
8	RECORD NO.		843
9	PAYMT ENCL		\$47.00
10			
11			
12			
13			
14			
15			
16			

	1	2
1234567890123456789012345678		

Message Enclosed

The receipt type MGEN is used for message enclosed transactions. The Diebold default receipt template is shown below, followed by a graphic example of the default receipt for a message from the cardholder to the institution placed in the envelope depository.

@RECORD NO. #SQ>
@MSG ENCL>>

1	12/15/96	08:30	S1AC9100
2			
3	376666000000000001-000		
4			
5	330 S. 108TH AVE		
6			
7			
8	RECORD NO.		843
9	MSG ENCL		
10			
11			
12			
13			
14			
15			
16			
17			
18			
19			

	1	2
1234567890123456789012345678		

Log Only

The receipt type OL is used for log only transactions. The Diebold default receipt template is shown below, followed by a graphic example of the default receipt for a log only transaction.

```
@RECORD NO.          #SQ>
@TYPE #OL @LOG ONLY MSG>>
```

1	12/15/96	08:30	S1AC9100
2			
3	376666000000000001-000		
4			
5	330 S. 108TH AVE		
6			
7			
8	RECORD NO.		843
9	TYPE 1 LOG ONLY MSG		
10			
11			
12			
13			
14			
15			
16			
17			
18			
19			

12

34567890123456789012345678

PIN Change

The receipt type PC is used for pin change transactions. The Diebold default receipt template is shown below, followed by a graphic example of the default receipt for a PIN change transaction.

@RECORD NO. #SQ>
@PIN CHANGE>>

1	12/15/96	08:30	S1AC9100
2			
3	376666000000000001-000		
4			
5	330 S. 108TH AVE		
6			
7			
8	RECORD NO.		843
9	PIN CHANGE		
10			
11			
12			
13			
14			
15			
16			
17			
18			
19			

1	2
1234567890123456789012345678	

Denied Financial Transactions

The Device Handler process converts receipt templates for denied transactions in the same manner as approved transactions with one exception. In addition to the receipt templates, a response code text table is used to associate one line of receipt text per ATM response code. Refer to the topic “Response Code Text Table for Financial Receipts” later in this section for a description of the response code text table.

The receipt type DENY is used for denied financial transactions. The Diebold default receipt template is shown below, followed by a graphic example of the default receipt for a denied withdrawal transaction.

```
@RECORD NO.      #SQ>
#TT>>
@TRANSACTION DECLINED>
#RC>
#RT>>
```

1	12/15/96	08:30	S1AC9100
2			
3	376666000000000001-000		
4			
5	330 S. 108TH AVE		
6			
7			
8	RECORD NO.	580	
9	WITHDRAWAL		
10	TRANSACTION DECLINED		
11	13		
12	EXPIRED CARD		
13			
14			
15			
16			
17			
18			
19			

12

1234567890123456789012345678

Response Code Text Table for Financial Receipts

The response code text table is a separate table in the C910NAMS (for 910 and 911 terminals) or C912NAMS (for 912 terminals) and is used in conjunction with the #RT (Response Text) data field when that particular data field is used in a receipt template for denied transactions.

The response code text table (RESPONSE^TEXT^G) provides additional receipt text for denied transaction receipts. This text gives the cardholder an explanation for the denied transaction and is in addition to the text provided in the receipt template.

The Device Handler process maps any BASE24 response code numbers that it receives into device-specific internal response code numbers. These internal response code numbers are what you will find in the C910NAMS (for 910 and 911 terminals) or C912NAMS (for 912 terminals). Refer to the ***BASE24-atm Transaction Processing Manual*** for complete descriptions of the BASE24-atm response codes.

Default entries for the response code text table are provided in the C910NAMS (for 910 and 911 terminals) or C912NAMS (for 912 terminals). The following table lists all of the default response code text messages in English. You can tailor all messages to meet your needs.

Device-specific Response Code	Default Response Code Text
1	INVALID PIN
2	PIN TRIES EXCEEDED
3	ENTER LESSER AMOUNT
9	TRANSACTION NOT ALLOWED
10	UNABLE TO PROCESS
11	INVALID TRAN OR ACCOUNT
12	UNAUTHORIZED CARD USAGE
13	EXPIRED CARD
14	INVALID CARD

Device-specific Response Code	Default Response Code Text
16	UNABLE TO PROCESS AMT
17	CREDIT CARD AMT
18	USES LIMIT EXCEEDED
19	INSUFFICIENT FUNDS
21	NO MATCHING SHARING GRP
22	CONTACT BANK
23	INSUFFICIENT FUNDS
34	UNABLE TO DISPENSE
35	UNABLE TO DEPOSIT
37	DAILY WD LIMIT REACHED
38	TRANSACTION TIMED OUT
41	UNABLE TO DISPENSE COINS
42	INVALID CASH BACK AMOUNT
44	NO CHECK ENTERED
45	UNABLE TO ACCEPT CHECKS

Mapping Response Codes to Receipt Text

The response code text table includes one line of receipt text for each response code. Each line of text can be configured in up to eight languages. Text for each language can be defined in groups of up to twenty-four characters each. The line of text from the table is printed on the denied transaction receipt at the location of the #RT field type code in the receipt template.

To locate the correct information in the response code text table, the Device Handler process performs the following steps:

1. Locates the applicable response code entry in the response code text table.

2. When a match is made on the response code, determines which portion of the entry to send to the ATM as receipt print data by using the language indicator. If the language indicator is 0, the Device Handler process uses the first twenty-four characters of the response code entry as receipt print data. If the language indicator is 1, the Device Handler process scans past the first twenty-four characters of the response code entry and uses the second group of twenty-four characters of the entry as receipt print data. Each additional language skips another twenty-four characters.

Administrative Transaction Receipts

The TDF accumulates transaction totals as transactions are completed for each ATM. These totals can then be printed on an ATM receipt. These totals are used in balancing the ATM and monitoring the cash supply. They can also be printed at the ATM via terminal balancing and terminal subtotalling administrative transactions. All balancing totals are accumulated from the last time the ATM was balanced and the totals were reset to zero.

Administrative Receipt Descriptions

Administrative transaction receipts are printed on the same type of receipt paper forms as those used for financial receipts.

The position of the information printed on the receipt is dependent on the width of the paper used and whether the paper form is printed or blank. The totals are printed in cassette descending order. That is, the totals for the cassette containing the highest denomination bills are printed first, while the totals for the lowest denomination cassette are printed last.

Formatting and contents of administrative receipts is wholly configurable according to your needs. Note that when establishing receipt templates, printer buffers have a maximum of 250 bytes. If additional information is to be sent to the ATM, an additional message is sent. The Administrative Header receipt template (ADHD) can be used to shorten header information on administrative receipts.

Diebold 910 and 911 administrative transactions print one transaction per receipt, while Diebold 912 administrative transactions can print one transaction per receipt, or one transaction may require two receipts, depending upon the number of cassettes in the terminal. Note that the Diebold 910 and 911 terminals use different field type codes than the Diebold 912 terminals.

Diebold 910/911 Administrative Transaction Field Type Code Descriptions

The following tables describe the standardized field type codes that are used in receipt templates for administrative receipts. Each field type code corresponds to a segment of fixed-length data. The data is retrieved from the TDF. Data names are shown in parentheses following each field type code description.

Cassette Field Type Code Descriptions

The following field type codes are used in that portion of the administrative receipt that includes cassette totals. Where applicable, TDF data names are included in parentheses after each field type code description.

Code	Description
HD	The number of bills removed from the high-bill cassette via midpoint adjustment during the current balancing period, multiplied by the denomination of the bills in the high-bill cassette (TDF.HOPR[1].CASH-DECR $\times$ TDF.HOPR[1].BILL-VAL).
HE	The current number of bills in the high-bill cassette, multiplied by the denomination of the bills in the high-bill cassette (TDF.HOPR[1].CASH-END $\times$ TDF.HOPR[1].BILL-VAL).
HI	The number of bills added to the high-bill cassette via midpoint adjustment during the current balancing period, multiplied by the denomination of the bills in the high-bill cassette (TDF.HOPR[1].CASH-INCR $\times$ TDF.HOPR[1].BILL-VAL).
HN	The hopper or cassette number.
HO	The number of bills dispensed from the high-bill cassette or removed from the high-bill cassette for cardholder withdrawals during the current balancing period, multiplied by the denomination of the bills in the high-bill cassette (TDF.HOPR[1].CASH-OUT $\times$ TDF.HOPR[1].BILL-VAL).
HS	The number of bills in the high-bill cassette at the beginning of the terminal's current balancing period, multiplied by the denomination of the bills in the high-bill cassette (TDF.HOPR[1].BEG-CASH $\times$ TDF.HOPR[1].BILL-VAL).
LD	The number of bills removed from the low-bill cassette via midpoint adjustment during the current balancing period, multiplied by the denomination of the bills in the low-bill cassette (TDF.HOPR[0].CASH-DECR $\times$ TDF.HOPR[0].BILL-VAL).
LE	The current number of bills in the low-bill cassette multiplied by the denomination of the bills in the low-bill cassette (TDF.HOPR[0].CASH-END $\times$ TDF.HOPR[0].BILL-VAL).

Code	Description
LI	The number of bills added to the low-bill cassette via midpoint adjustment during the current balancing period, multiplied by the denomination of the bills in the low-bill cassette (TDF.HOPR[0].CASH-INCR $\times$ TDF.HOPR[0].BILL-VAL).
LO	The number of bills dispensed from the low-bill cassette or removed from the low-bill cassette for cardholder withdrawals during the current balancing period, multiplied by the denomination of the bills in the low-bill cassette (TDF.HOPR[0].CASH-OUT $\times$ TDF.HOPR[0].BILL-VAL).
LS	The number of bills in the low-bill cassette at the beginning of the terminal's current balancing period, multiplied by the denomination of the bills in the low-bill cassette (TDF.HOPR[0].BEGCASH $\times$ TDF.HOPR[0].BILL-VAL).

Each cassette field type code can be followed with a variable length monetary symbol in the receipt template to identify the monetary unit used for the transaction.

Totals for a specific cassette are printed only if the bill value for that cassette has been defined in the TDF.

Depository Field Type Code Descriptions

The following field type codes are used in the portion of the administrative receipt that includes depository totals.

Code	Description
AC	The total unverified amount of commercial deposits accepted since the terminal was last balanced. (TDF.AMT-CMRCL-DEP)
AD	The unverified amount of deposits accepted since the terminal was last balanced. (TDF.AMT-DEP)
AK	The unverified amount of checks cashed or deposited and received since the terminal was last balanced. (TDF.AMT-CHK)

Code	Description
AP	The unverified amount of payments enclosed received since the terminal was last balanced. (TDF.AMT-PAY)
AT	Administrative type (for example, SUB for subtotals). The text that is printed is in the SUB^TOT^TBL^G table in the C910NAMS.
CK	The total number of checks cashed or deposited since the terminal was last balanced. (TDF.NUM-CHK)
CR	The total number of cards retained since the terminal was last balanced. (TDF.CRDS-RET)
D0	The number of deposits received since the terminal was last balanced. (TDF.NUM-DEP)
D1	The number of commercial deposits received since the terminal was last balanced. (TDF.NUM-CMRCL-DEP)
EN	The sum of the following: number of deposits, number of commercial deposits, number of message envelopes, number of payment envelopes, and number of checks cashed or deposited since the terminal was last balanced.
MG	The total number of “message to financial institution” envelopes received since the terminal was last balanced. (TDF.NUM-MSG)
PY	The total number of payment envelopes received since the terminal was last balanced. (TDF.NUM-PAY)
RA	Response text for denied administrative transactions. Text table is in the C910NAMS for 910 and 911 terminals.

Note: Each depository amount field type code can be followed with a variable length monetary symbol, which identifies the monetary unit used for the transaction.

Diebold 910/911 Administrative Transaction Templates

There are two administrative receipt templates used for approved administrative transactions: ADHO for cassette information and ADTO for administrative totals. In addition, ADDN is a template used for denied administrative transactions.

The administrative header is receipt type ADHD, and can be used to shorten the standard header so more administrative information can fit on one receipt. The administrative header does not have a default receipt template in the RCPT. If you create a receipt type ADHD, this receipt type will only be used for administrative transactions. If the ADHD receipt type does not exist, the HDR receipt type will be used for financial and administrative transactions.

The following are representations of the default entries in the RCPT for administrative receipts.

Transaction Type	Receipt Type	Default RCPT Template Data
Administrative transaction denied	ADDN	@ADMIN TRANSACTION DENIED> #RA>>
Cassette Totals	ADHO	@HB STR#HS\$> @HB INC#HI\$> @HB DEC#HD\$> @HB OUT#HO\$> @HB END#HE\$> @LB STR#LS\$> @LB INC#LI\$> @LB DEC#LD\$> @LB OUT#LO\$> @LB END#LE\$>
Depository Totals (for terminal subtotaling, terminal balancing, and midpoint adjustment	ADTO	@DEP0#D0> @DEP1#D1> @MSG #MG> @PAY #PY> @TOT #EN> @CRD #CD #AT

Diebold 910/911 Sample Administrative Receipt

The following is an example of a Diebold ATM default receipt format for an administrative subtotal transaction. Note that this same format is used for all administrative transactions

```

1      12/15/96  08:30   S1AC9100
2
3      376666000000000001-000
4
5      330 S. 108TH AVE
6
7
8      CO STR $10000
9      CO INC      $0 CO DEC      $0
10     CO OUT      $0 CO END $10000
11     C1 STR $10000
12     C1 INC      $0 C1 DEC      $0
13     C1 OUT      $0 C1 END $10000
14     DEP0 00
15     DEP1 00
16     MSG  00
17     PAY  00
18     TOT  00
19     CRD  00 SUB
20

```

1 2
1234567890123456789012345678

Diebold 912 Administrative Receipts

The Diebold 912 Device Handler process requires its own set of field type codes and receipt templates for administrative receipts.

The administrative receipt for the Diebold 912 terminal can be split into two sections, or printed on two receipts. If the terminal has less than four cassettes, the cassette totals are on the top half of the receipt, and the depository totals are on the

bottom half of the receipt. If the terminal has four cassettes, the cassette totals are printed on the first receipt, while the depository totals are printed on the second receipt.

There is one receipt group table and one or more receipt templates for each half of the administrative receipt. The Device Handler process converts receipt templates for administrative transactions in the same manner as approved and denied financial transactions. However, administrative receipts require their own set of field type codes for the receipt templates. The field markers used are the same as those used for approved and denied financial transactions for the Diebold 910, 911, and 912 terminals. These field markers are described earlier in this section.

The Diebold 912 transaction record printer can handle 26 characters per line and 20 lines per receipt. These restrictions must be considered when creating receipt templates.

Diebold 912 Administrative Transaction Field Type Codes

The following tables describe the standardized field type codes that are used in receipt templates for administrative receipts. Each field type code corresponds to a predetermined piece of fixed-length data. The data is retrieved from the Terminal Data File (TDF). Data names are shown in parentheses following each field type code description.

Cassette Field Type Code Descriptions

The following field type codes are used in that portion of the administrative receipt that includes cassette totals. Where applicable, TDF data names are included in parentheses after each field type code description. Further explanation of TDF data names is available using DDLDOC. Refer to the ***BASE24 Files and Record Structures Reference Manual*** for DDLDOC operating instructions.

Code	Description
BV	The denomination of the bills in the cassette (TDF.HOPR[<i>i</i>].BILL-VAL where <i>i</i> refers to cassettes 0 through 3 for Diebold ATMs).

Code	Description
CD	The number of bills removed from the cassette via midpoint adjustment during the current balancing period multiplied by the denomination of the bills in the cassette. This field type code has a 10-byte length. (TDF.HOPR[<i>i</i>].CASH-DECR $\times$ TDF.HOPR[<i>i</i>].BILL-VAL where <i>i</i> refers to cassettes 0 through 3 for Diebold ATMs.)
CE	The current number of bills in the cassette multiplied by the denomination of the bills in the cassette. This field type code has a 10-byte length. (TDF.HOPR[<i>i</i>].CASH-END $\times$ TDF.HOPR[<i>i</i>].BILL-VAL where <i>i</i> refers to cassettes 0 through 3 for Diebold ATMs.)
CI	The number of bills added to the cassette via midpoint adjustment during the current balancing period multiplied by the denomination of the bills in the cassette. This field type code has a 10-byte length. (TDF.HOPR[<i>i</i>].CASH-INCR $\times$ TDF.HOPR[<i>i</i>].BILL-VAL where <i>i</i> refers to cassettes 0 through 3 for Diebold ATMs.)
CO	The number of bills dispensed from the cassette or removed from the cassette for cardholder withdrawals during the current balancing period multiplied by the denomination of the bills in the cassette. This field type code has a 10-byte length. (TDF.HOPR[<i>i</i>].CASH-OUT $\times$ TDF.HOPR[<i>i</i>].BILL-VAL where <i>i</i> refers to cassettes 0 through 3 for Diebold ATMs.)
CS	The number of bills in the cassette at the beginning of the current balancing period multiplied by the denomination of the bills in the cassette. This field type code has a 10-byte length. (TDF.HOPR[<i>i</i>].BEG-CASH $\times$ TDF.HOPR[<i>i</i>].BILL-VAL where <i>i</i> refers to cassettes 0 through 3 for Diebold ATMs.)
HN	The hopper or cassette number.

Each cassette field type code must be followed with a variable-length monetary symbol in the receipt template to identify the monetary unit used for the transaction.

Totals for a specific cassette are printed only if the bill value for that cassette has been defined in the Terminal Data File (TDF).

Depository Field Type Code Descriptions

The following field type codes are used in the portion of the administrative receipt for the Diebold 912 terminal that includes depository totals. TDF data names are included in parentheses. Further explanation of TDF data names is available using DDLDOC. Refer to the ***BASE24 Files and Record Structures Reference Manual*** for DDLDOC operating instructions.

Code	Description
AC	The total unverified amount of commercial deposits accepted since the terminal was last balanced. (TDF.AMT-CMRCL-DEP)
AD	The unverified amount of deposits accepted since the terminal was last balanced. (TDF.AMT-DEP)
AK	The unverified amount of checks cashed or deposited and received since the terminal was last balanced. (TDF.AMT-CHK)
AP	The unverified amount of payments enclosed received since the terminal was last balanced. (TDF.AMT-PAY)
AT	Administrative type (for example, SUB for subtotals). The text that is printed is in the DIEBOLD^912^SUB^TOT^TBL table in the C912NAMS.
CK	The total number of checks cashed or deposited since the terminal was last balanced. (TDF.NUM-CHK)
CR	The total number of cards retained since the terminal was last balanced. (TDF.CRDS-RET)
D0	The number of deposits received since the terminal was last balanced. (TDF.NUM-DEP)
D1	The number of commercial deposits received since the terminal was last balanced. (TDF.NUM-CMRCL-DEP)
EN	The sum of the following: number of deposits, number of commercial deposits, number of message envelopes, number of payment envelopes, and number of checks cashed or deposited since the terminal was last balanced.

Code	Description
MG	The total number of “message to financial institution” envelopes received since the terminal was last balanced. (TDF.NUM-MSG)
PY	The total number of payment envelopes received since the terminal was last balanced. (TDF.NUM-PAY)
RA	Response text for denied administrative transactions. Text table is in the or C912NAMS for 912 terminals.

Each depository amount field type code must be followed with a variable-length monetary symbol, which identifies the monetary unit used for the transaction.

Diebold 912 Administrative Transaction Templates

There are two administrative receipt templates used for approved administrative transactions: ADHO for cassette information and ADTO for administrative totals. In addition, ADDN is a template used for denied administrative transactions.

The administrative header is receipt type ADHD, and can be used to shorten the standard header so more administrative information can fit on one receipt. The administrative header does not have a default receipt template in the RCPT. If you create a receipt type ADHD, this receipt type will only be used for administrative transactions. If the ADHD receipt type does not exist, the HDR receipt type will be used for financial and administrative transactions.

The following are representations of the default entries in the RCPT for administrative receipts.

Transaction Type	Receipt Type	Default RCPT Template Data
Administrative transaction denied	ADDN	@ADMIN TRANSACTION DENIED> #RA>>
Cassette Totals	ADHO	#BV@ S#CS\$> @I#CI\$@ D#CD\$> @O#CO\$@ E#CE\$>

Transaction Type	Receipt Type	Default RCPT Template Data
Depository Totals (for terminal subtotaling, terminal balancing, and midpoint adjustment)	ADTO	@DEP0#D0> @DEP1#D1> @MSG #MG> @PAY #PY> @TOT #EN> @CRD #CR #AT>

Diebold 912 Sample Administrative Receipt

The following is an example of a Diebold ATM default receipt format for an administrative subtotal transaction. Note that this same format is used for all administrative transactions.

1	12/15/96	08:30	S1AC9100
2			
3	376666000000000001-000		
4			
5	330 S. 108TH AVE		
6			
7			
8	20	S	\$10000
9	I	\$0	D \$0
10	0	\$0	E \$10000
11	10	S	\$5000
12	I	\$0	D \$0
13	0	\$20	E \$4980
14	DEP0 00	DEP1 00	
15	MSG 00	PAY 00	
16	TOT 00	CRD 00	SUB
17			
18			
19			
20			

1
2

1234567890123456789012345678

Response Code Text Table for Administrative Receipts for All Diebold Terminals

The denied administrative transaction response code text table is a separate table in the C910NAMS (for 910 and 911 terminals) or C912NAMS (for 912 terminals) and is used in conjunction with the #RA (Response Text Administrative) field type code when that particular field type code is used in an administrative receipt template for denied transactions.

The response code text table for denied administrative transactions (ADMIN^RESP^TEXT^G) provides additional receipt text for denied transactions. This text gives the service person an explanation for the denied transaction and is in addition to the text provided in the receipt template.

When the Device Handler process encounters a #RA field type code in a receipt template, the process takes the response code that it generated or that it received from the Authorization process, maps that code to an internal response code, moves to the administrative response code text table and scans for a match on the internal response code. From this point on, the Device Handler process uses the administrative response code text table in the same manner as the response code text table for financial transactions. This processing is described earlier in this section. The following entries are the default entries in the response code text table for denied administrative transactions. All of the default entries are in English; however, up to eight languages can be defined for the Diebold terminals.

Response Code	Initial Message Text
16	UNABLE TO PROCESS AMT
29	INVALID AMT FOR HOPPER
33	ATM ALREADY BALANCED
39	ADMIN TRAN CANCELLED

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Appendix A

Philips Enhanced Display Screens

This appendix explains how to access the enhanced display entry screens that are used to modify the Configuration File for Diebold 912 ATMs equipped with the Philips enhanced character generator programmable read only memory (PROM). Descriptions are also provided for each of the enhanced display screens that are used to modify the ATM Configuration File (CNFG912).

Enhanced Display Screens

Enhanced display definition screens are used to set up configuration data allowing special character sets (libraries) to be accessed, or created, and used on the ATM's display screens.

There are two types of enhanced character PROMs available for Diebold 912 ATMs. The MDS enhanced character generator PROM contains predefined character libraries. With this PROM, complete library definitions do not need to be downloaded with the exception of multiblock character sets.

The Philips enhanced character generator PROM allows you to create and use customized libraries. Using this PROM, all of the information necessary to define the libraries and the characters available within those libraries must be downloaded.

This appendix provides documentation for the enhanced display entry screens used to modify the Configuration File for Diebold 912 ATMs equipped with the Philips enhanced character generator PROM. For information on the enhanced display configuration screens used to modify the Configuration File for ATMs equipped with an MDS enhanced character generator PROM, refer to section 4.

Accessing the Enhanced Display Screens

The Diebold entry program (ENT9000) is used to access the enhanced display screens that are used to modify the Configuration File (CNFG912) for Diebold 912 ATMs that contain the Philips enhanced character generator PROM. When the ATM contains the Philips PROM, the file code for the Configuration File must be changed to access the appropriate enhanced display screens.

The file code for the Configuration File can be changed from the TACL prompt by entering the following:

```
FUP ALTER CNFG912, CODE 0
```

After the file code is changed to 0, the entry program is accessed in the usual way (that is, a run command is issued from the TACL prompt). When the entry program is accessed, the system displays the menu for the Diebold entry program. This menu is used only to select the enhanced display data to be accessed and modified.

Configuration File Function Keys

The function keys specific to most of the screens associated with the Diebold 912 Configuration File are described below.

The first column of information below shows the BASE24 keys. The second column describes the functions that can be accomplished with these function keys.

Key	Description
F1	ENTER — Checks the data that has been entered on the screen for errors and enters the data in the file.
F2	READ DATA — Reads data in the file.
F3	ADD DATA — Adds data to the file. The data added must be unique within the file.
F4	DELETE DATA — Deletes data from the file.
F5	UPDATE DATA — Changes data already in the file.
F9	NEXT — Displays the next screen of the file in which you are working. If you are on the last screen of the file, this function key takes you to the first screen of the file.
F10	PRINT SCREEN — Sends the screen currently being displayed to the spooler location specified on the Diebold Configurator Process menu in the OUTPUT DEVICE FOR PRINT FUNCTION field as explained in this appendix.
F16	EXIT — Exits the Diebold entry program or the configuration screen currently being accessed. This function key is operable on the Help screen to enable you to resume normal processing.

Diebold Configurator Process Menu

The following Diebold Configurator Process menu displays the functions available for maintaining the Diebold 912 Configuration File (CNFG912).

```

                                DIEBOLD CONFIGURATOR PROCESS - VERSION 2.33 (25Aug93)

Code  Functional Description
-----
  1.  ATM Display Screens    ***> PRINT functions from this menu generate cross-
  2.  State Tables          ***> references of Displays, Overlays, and States.
  3.  Miscellaneous Data
  4.  Pin Data
  5.  FIT (Financial Institution Table)
  6.  Display/Alter Configuration Mode
  7.  Enhanced Display Library Definition    ***> Enhanced Displays and Standard
  8.  Enhanced Display Character Definition ***> Displays are mutually-exclusive
  9.  Enhanced Display Screen Definition    ***> and depend on ATM's strapping.

Enter Function Code (from above menu) and press appropriate function key ...

Diebold Configuration File Name ..... $REL5.AT50C912.CNFG912
Output Device for PRINT function ..... $S.#HOLD

=====
F1=ENTER      F3=CREATE FILE      F10=PRINT      F16=EXIT
                SHIFT-F10=PRINT FILE

```


Field Descriptions

The following fields are used when accessing the Diebold Configurator Process screen.

Enter Function Code — A code identifying the configuration data to be modified. This function code is used to select enhanced display configuration data to be modified. The **F1** key is pressed to enter the desired code. If a code between 7 and 9 is entered, the system responds by displaying the appropriate enhanced display screen for modifying the enhanced display data. The following codes are discussed in this appendix:

- 7 = Enhanced Display Library Definition. This code allows you to access the Diebold Enhanced Display Library Definition screen.
- 8 = Enhanced Display Character Definition. This code allows you to access the Enhanced Display Character Definition screen.
- 9 = Enhanced Display Screen Definition. This code allows you to access the Enhanced Display Screen Definition screen.

Refer to section 4 for a complete list of the function codes.

Diebold Configuration File Name — The file name of the Diebold Configuration File. This name must be entered using the Tandem file naming convention of `$volume.subvolume.CNFG912`.

When the file name entered does not exist, the system displays a message at the bottom of the screen advising that the file name entered does not exist. You then select another file, or press the **F3** key.

To create a new Configuration File, the new file name must be entered directly over the displayed file name and the **F3** key must be pressed. The system responds by creating the new file and displays a message at the bottom of the screen indicating that the new file was created.

At times, the file can be opened with READ-ONLY access due to a possible security violation or because another process has the CNFG912 file open. The message, "FILE CAN ONLY BE OPENED READ-ONLY" is displayed at the bottom of the screen. When this occurs, all functions can still be performed except for add, delete, or update.

You should check for a possible security violation. If another process has the file open, you must select another file or wait until the other user is finished.

Output Device for Print Function — The name of the output device for items to be printed.

This field is required and can be either a printer (or the Spooler), an entry-sequenced file, or another terminal. It cannot, however, be the same terminal where the Diebold entry program is running.

When the output device named is another terminal, the Diebold entry program performs paging and prompts you to press the **Return** key before starting a new page. To stop the print function, press the **Control-Y** keys simultaneously.

Whenever a print function is in progress, the terminal keyboard is locked (all keyboard functions are disabled). Once the print function is completed, the keyboard is unlocked.

Function Keys

The use of one function key on the Diebold Configurator Process screen varies from the standard function keys explained earlier in this appendix. In addition, one key which was not assigned a function in the standard definitions is assigned a special function for this screen. The use of these keys is explained below.

The first column of information below shows the BASE24 keys. The second column describes the functions that can be accomplished with these function keys.

Key	Description
F3	CREATE FILE — Creates a new Configuration File.
Shift-F10	PRINT FILE — Prints the contents of the Configuration File to specified output device.

Enhanced Display Library Definition Screen

The Enhanced Display Library Definition screen is used with Diebold 912 ATMs using the Philips enhanced character generator PROM.

The Enhanced Display Library Definition screen is used to define a library, or character set, controlling the size of the character set (that is, how many rows and columns of pixels are used to define a character), margins on the ATM's CRT, etc.

At present the following language is defined. Up to nine libraries can be defined as follows:

Library	Language
1	English
2–9	User-defined

For further information on the enhanced display feature including field descriptions for this screen, refer to ***Diebold TABS Manual, #386***.

The Enhanced Display Library Definition screen is shown below.

ENHANCED DISPLAY
LIBRARY DEFINITION

Library 1

Top Screen Margin 000
Left Screen Margin 000

Character Block Height 015
Character Block Width 010

Top Character Margin 004
Left Character Margin 002

Character Height 009
Character Width 008

Character Rows per Screen 016
Characters per Row 032

F2=READ
F9=NEXT
F3=ADD
F10=PRINT
F4=DELETE
F5=UPDATE
F16=EXIT

Enhanced Display Character Definition Screen

The Enhanced Display Character Definition screen can be used with Diebold 912 ATMs using the Philips enhanced character generator PROM.

The Enhanced Display Character Definition screen allows for designing and modifying each character in a library. There are two screens available for this purpose. The high-level Enhanced Display Character Definition screen displays a character that is enlarged so that each pixel is one character on the screen. With this screen, the character can be drawn as it will appear on the ATM screen.

The low-level Enhanced Display Character Definition screen is available by pressing the **F1** key from the high-level screen. The low-level screen allows the character definition to be edited in its native format, which is the hexadecimal representation of the character, treating each pixel as a bit that is either ON or OFF. Each character is then referenced by its index into the library. The default library indexes are, in part: A—Index 21, B—Index 22, ... Y—Index 39, and Z—Index 3. Note that Index 21 starts with A to support Track 1 read. Up to 240 characters can be defined per library.

The high-level and low-level Enhanced Display Character Definition screens are shown by the following illustrations.

```

Character 00 Group 0000      ....+....1....+....2....+....3....+....4
Library Identifier 1         1 >
                             2 >
*****                      3 >
*   ENHANCED DISPLAY   *    4 >
* CHARACTER DEFINITION *    5 >
*****                      6 >
                             7 >
+-----+                  8 >
| Using blanks and      | 9 >
| asterisks, "draw" the | 10 >
| character as it should | 11 >
| appear on the CRT.    | 12 >
|                       | 13 >
| Make sure the height  | 14 >
| and width you use to  | 15 >
| draw the character    | 16 >
| match those defined in | 17 >
| the Enhanced Display  | 18 >
| Library Definition.   | 19 >
+-----+                20 >
                             ....+....1....+....2....+....3....+....4

```

```

      ENHANCED DISPLAY CHARACTER DEFINITION - LOW LEVEL

      Character 00  Group 0000  Library 1

      Enter a string of two-digit hex-graphic numbers which define the desired
      character, remembering that it requires ( (CHAR HEIGHT/8) * CHAR WIDTH )
      hex-graphic numbers to define the character.

      1- 10:
      11- 20:
      21- 30:
      31- 40:
      41- 50:
      51- 60:
      61- 70:
      71- 80:
      81- 90:
      91-100:

=====
      F1=HI - LEVEL      F2=READ      F3=ADD      F4=DELETE      F5=UPDATE

```

Function Keys

The use of one function key on the Diebold Configurator Process screen varies from the standard function keys explained earlier in this appendix. The use of this key is explained below.

The first column of information below shows the BASE24 key. The second column describes the function that can be accomplished with this function key.

Key	Description
F1	LO-LEVEL/HI-LEVEL — Displays the low-level or the high-level screen. When this function key is pressed from the high-level screen, the low-level screen is displayed. When this function key is pressed from the low-level screen, the high-level screen is displayed.

Enhanced Display Screen Definition Screen

The Enhanced Display Screen Definition screen can be used with Diebold 912 ATMs using the Philips enhanced character generator PROM.

This screen is used to define each screen to be displayed by the ATM. Due to the limitation of the terminal on which the Diebold entry program (ENT9000) is running, this function must be performed at the native command level. Therefore, caution should be used when modifying enhanced displays, since the actual appearance of the screen when it is displayed by the Diebold ATM is not shown by the Diebold entry program.

The Enhanced Display Screen Definition screen is illustrated below.

ENHANCED DISPLAY SCREEN DEFINITION Screen Number 000 Group 0000 Enter screen commands below, separating each command and each byte of data with a comma. Command syntaces are at right.		<pre> +-----+ FF=Clear Screen HT=Track1 Name SI=Set Cursor , <row> , <col> S0=Overlay Screen , <screen #> VT=Use Library , <library #> xx=Character Index +-----+ </pre>
<pre>1.....2.....3.....4.....5.....6.....7.....+ 1 2 3 4 5 6 7 8 9 101.....2.....3.....4.....5.....6.....7.....+ </pre>		
<pre> ===== F2=READ F3=ADD F4=DELETE F5=UPDATE </pre>		

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